

Insights from the Pareto/NBD synthetic grid studies

Stored forecasts under `Studies/seasonal_4x4x10*` .

Every test follows `docs/statistical-protocol.md` : Δ is the difference of means with a 95% percentile-bootstrap interval (`evaluation.effects.effect`), run per rate $\times$ churn cell only (paired by panel; independent for 1,000 vs 3,000 customers).

How the datasets were made

Every panel was simulated from a known Pareto/NBD process, so the true model is known and Pareto/NBD is the correct model by construction. That makes it the ceiling the neural models are measured against.

The process, per customer (`data_preparation/pareto_nbd_simulation.py`):

- Purchase rate $\lambda \sim \text{Gamma}(r, \alpha)$. While alive, weekly purchases are $\text{Poisson}(\lambda)$.
- Dropout rate $\mu \sim \text{Gamma}(s, \beta)$. Lifetime $\tau \sim \text{Exponential}(\mu)$. After τ the customer never buys again.
- A fixed seasonal multiplier scales every customer's weekly rate: peaks at weeks 12, 25, 30 and 47, amplitude 1.5, width 3 weeks. It averages to 1 over the year, so the annual rate is unchanged.

The grid (`grids/seasonal_4x4x10.py`): two axes, set in plain terms and turned into α and β by the generator.

Setting	Values
Mean weekly purchase rate (r/α)	0.01, 0.05, 0.10, 0.30
Churn rate: share of customers dropped out by week 52	20%, 40%, 60%, 80%
Shape parameters	$r = s = 2.0$
Replicate panels per cell	10 (different seeds, base seed 42)
Total panels (datasets)	$4 \times 4 \times 10 = 160$
Customers per panel	1,000 (a second grid, <code>seasonal_4x4x10_n3000</code> , uses 3,000)
Length	156 weeks, 1999–2001

Windows. 1999 trains, 2000 is the validation window (the temporal split of ADR-0001), 2001 is the holdout. Customers with no purchase in calibration are dropped, as in a real cohort. Weekly counts are capped at 6, which gives the models a 7-class softmax head.

What the models see. In the baseline arm, only each customer's weekly count plus the calendar (year index and week sin/cos). There are no covariates; the other arms add encodings of the customer's own history (next section). The true λ , μ and τ are saved per customer but never shown to a model.

What was trained and how it was scored

Three models ran on all 160 panels: an LSTM, a Transformer and the Pareto/NBD benchmark. The frozen ValentinLSTM did not run on this grid. It reads only embedded columns (count and week), and every arm here, the `no_ar` (no autoregressive features) baseline included, carries the continuous calendar columns (year index, week sin/cos), which it refuses (F11). It did run without covariates on the real panels (CDNOW, electronics), where the calendar can be left out.

Architectures and search (grids/seasonal_4x4x10.py). Each neural study is one 100-trial Optuna search on one panel, then a refit and a 200-path Monte Carlo forecast.

Model	Searched hyperparameters	Training
LSTM	batch {32, 64, 128, 256}; learning rate 1e-4–1e-2 (log); embedding {64, 128, 256}; hidden {32, 64, 128}; dense {32, 64, 128}; dropout {0, 0.2, 0.4}	up to 100 epochs, patience 7, cross-entropy
Transformer	batch {32, 64, 128, 256}; d_model {32, 64, 128}; heads {2, 4, 8}; layers 1–3; dropout {0, 0.1, 0.2, 0.3}; learning rate 1e-4–3e-3 (log)	same
Pareto/NBD	none: one hierarchical-Bayes MCMC fit (BTYDplus defaults)	—

Both neural models use the `valendin` (original paper) embedder. The `projected` embedder was declared but cut on cost.

Arms: how a customer's history is fed in. Three AR encodings crossed with two cluster settings = 6 arms per neural model, 12 neural trees plus Pareto/NBD, 2,080 studies ([vast.ai](#), 4–6 September 2026).

Arm part	Value	What the model gets
AR encoding	no_ar	count and calendar only (the baseline)
	ar_unbounded	recency, frequency and age as counters: Pareto/NBD's own sufficient statistic (t_x , x , T)
	ar_bounded	the same history as 0/1 flags: active in the last 2 / 4 / 8 / 16 / 32 weeks, plus has-bought-before
Cluster	no_cluster / kmeans_8	none, or a fixed k-means label ($K = 8$) on (t_x , x , T)

A second grid with 3,000 customers per panel (17–18 September) re-ran two LSTM arms (`no_ar` and `ar_bounded` , no cluster) and Pareto/NBD. The Transformer was not run there.

Metrics. One study per panel, so every spread below is across panels, not training runs.

Metric	What it measures	Good value
RMSE (customer totals)	error in each customer's holdout-year total	low
Bias %	total forecast vs total actual over the holdout year	0
MAPE (aggregate)	error in the weekly total, week by week	low
Win rate	share of panels where a model's MAPE beats Pareto/NBD's on the same panel	high
Spearman	rank correlation of each customer's predicted and actual holdout-year total, over the panel's customers	1
Validation CE	cross-entropy per customer-week of the chosen trial on the 2000 validation window (the Optuna objective, one step ahead with the true counts fed in). Neural models only: Pareto/NBD emits no class distribution. Depends on the panel's rate and churn, so compare within one cell	low

The per-cell Results tables show five metrics, each as the mean over the cell's panels with its 95% percentile-bootstrap interval in brackets; tables that pool cells, and the later sections, show **RMSE on customer totals / bias % / MAPE** as means only. RMSE is computed on each customer's holdout-year total, the paper's definition. It grows with the purchase rate (about 0.6 at rate 0.01, about 5 at rate 0.30), so it is compared within one rate only. The per-week RMSE in `results.csv` is not used: 10 of 13 trees score 0.18 on it.

Tests (docs/statistical-protocol.md). One panel is one replication, and every test runs inside one rate × churn cell of 10 panels. The effect is the difference of means, $\Delta = \text{mean}(B) - \text{mean}(A)$, with its 95% percentile-bootstrap interval from 10,000 resamples

(panelclv.evaluation.effects.effect), and a claim is supported when the interval excludes 0. Two trees fitted on the same panels are paired by panel. The 1,000- and 3,000-customer grids are independent replications.

Results

The ranking depends on both the purchase rate and the churn rate. On MAPE, Pareto/NBD leads every cell at rates 0.01–0.05 and at rate 0.10 with churn 20–40%; the LSTM with bounded flags leads at rate 0.10 with churn 60–80% and at rate 0.30 with churn 20–40%, and the LSTM without AR features at rate 0.30 with churn 60–80%. The Transformer leads no cell. On Spearman, Pareto/NBD leads all 16 cells, and in every one its lead over the runner-up is supported. Two Transformer ar_unbounded studies whose forecasts did not match their stored results are left out.

Setting for this section. Seasonal panels (4 peaks, amplitude 1.5), 1,000 customers, 13 trees (Pareto/NBD and 6 arms each of the LSTM and the Transformer). Four purchase rates × four churn rates (20, 40, 60 and 80% of customers dropped out by week 52), 10 panels per cell. A metric entry in a per-cell table is the mean over the cell's panels with its 95% percentile-bootstrap interval. RMSE is on customer totals and grows with the rate, and CE depends on the panel, so compare both within one table only. The three "Beats P/NBD" columns compare the tree with Pareto/NBD on the same panel, one metric each, and count the table's panels where the tree wins: lower MAPE, smaller |bias| (closer to 0, either sign), higher Spearman. The counts are description only and carry no marks.

Best tree (per-cell tables only): Every metric column of a cell's table marks its best tree in **bold**: the lowest mean over the cell's panels for RMSE, MAPE, Val. CE and |bias| (taken per panel, so a tree that errs +50% on one panel and −50% on another is not unbiased), and the highest mean for Spearman. A tree is underlined when the paired 95% bootstrap interval of its difference from the best tree, over the cell's 10 panels, contains 0. Read an underline as "no clear difference from the best at n = 10", never as "equal".

By purchase rate and churn

One table per rate × churn cell, 10 panels per row (9 where a study was left out).

Rate 0.01, churn 20% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	0.79 [0.77, 0.83]	+12 [+9, +15]	48 [45, 51]	0.22 [0.19, 0.25]	—	—	—	—
LSTM	no_ar	0.87 [0.83, 0.91]	+59 [+47, +74]	71 [62, 82]	0.01 [−0.02, 0.04]	0.079 [0.076, 0.082]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded	6.70 [1.95, 12.65]	+502 [+93, +1058]	513 [104, 1069]	−0.16 [−0.20, −0.12]	0.078 [0.074, 0.081]	1 of 10	1 of 10	0 of 10
LSTM	ar_bounded	0.86 [0.82, 0.92]	+44 [+23, +67]	65 [53, 81]	−0.03 [−0.09, 0.02]	0.078 [0.075, 0.081]	3 of 10	1 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	0.88 [0.85, 0.92]	+56 [+38, +71]	72 [62, 82]	0.13 [0.10, 0.17]	0.077 [0.074, 0.080]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	5.09 [2.65, 7.70]	+235 [+107, +377]	250 [124, 390]	−0.08 [−0.13, −0.04]	0.075 [0.072, 0.078]	0 of 10	0 of 10	0 of 10
LSTM	no_ar + kmeans_8	0.91 [0.87, 0.97]	+77 [+56, +98]	86 [70, 104]	0.14 [0.11, 0.16]	0.078 [0.075, 0.081]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	0.84 [0.80, 0.90]	+19 [−6, +45]	58 [50, 68]	0.05 [0.03, 0.08]	0.077 [0.074, 0.081]	3 of 10	3 of 10	0 of 10
Transformer	ar_unbounded	0.83 [0.81, 0.87]	+7 [−9, +26]	49 [44, 56]	−0.05 [−0.09, 0.00]	0.077 [0.074, 0.080]	6 of 10	6 of 10	0 of 10
Transformer	ar_bounded	0.85 [0.81, 0.91]	+36 [+11, +62]	62 [50, 79]	0.03 [0.00, 0.06]	0.077 [0.074, 0.080]	3 of 10	1 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	0.90 [0.84, 0.98]	+23 [−17, +72]	73 [55, 102]	0.16 [0.14, 0.19]	0.074 [0.071, 0.078]	2 of 10	0 of 10	2 of 10
Transformer	ar_unbounded + kmeans_8	0.87 [0.83, 0.92]	−13 [−42, +21]	65 [57, 73]	0.03 [−0.03, 0.08]	0.072 [0.068, 0.076]	0 of 10	2 of 10	0 of 10
Transformer	no_ar + kmeans_8	0.86 [0.81, 0.92]	−3 [−27, +22]	55 [49, 63]	0.12 [0.09, 0.16]	0.077 [0.074, 0.080]	4 of 10	2 of 10	0 of 10

Rate 0.01, churn 40% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	0.68 [0.65, 0.71]	+34 [+26, +40]	66 [62, 68]	0.25 [0.23, 0.28]	—	—	—	—
LSTM	no_ar	0.80 [0.77, 0.83]	+132 [+114, +150]	136 [120, 154]	0.01 [−0.02, 0.04]	0.068 [0.065, 0.071]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded	1.89 [0.82, 3.37]	+182 [+123, +251]	189 [130, 257]	−0.23 [−0.25, −0.21]	0.068 [0.064, 0.071]	0 of 10	0 of 10	0 of 10
LSTM	ar_bounded	0.78 [0.75, 0.82]	+114 [+100, +130]	121 [108, 135]	−0.09 [−0.12, −0.06]	0.067 [0.064, 0.070]	0 of 10	0 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	0.80 [0.76, 0.83]	+116 [+99, +132]	121 [107, 136]	0.11 [0.07, 0.14]	0.067 [0.064, 0.070]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	11.37 [5.39, 17.90]	+1300 [+412, +2406]	1304 [418, 2409]	−0.12 [−0.19, −0.05]	0.066 [0.063, 0.069]	0 of 10	0 of 10	0 of 10
LSTM	no_ar + kmeans_8	0.81 [0.78, 0.84]	+136 [+123, +149]	140 [128, 152]	0.14 [0.11, 0.16]	0.067 [0.064, 0.070]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	0.75 [0.69, 0.82]	+61 [+13, +113]	96 [65, 134]	0.02 [−0.03, 0.06]	0.066 [0.063, 0.068]	6 of 10	6 of 10	0 of 10
Transformer	ar_unbounded	0.72 [0.69, 0.75]	+18 [−21, +63]	78 [59, 100]	0.00 [−0.07, 0.08]	0.065 [0.062, 0.068]	6 of 10	5 of 10	1 of 10
Transformer	ar_bounded	0.71 [0.69, 0.74]	+27 [−9, +61]	75 [65, 87]	−0.02 [−0.04, 0.00]	0.065 [0.062, 0.068]	4 of 10	2 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	0.80 [0.70, 0.92]	+64 [+11, +126]	102 [66, 149]	0.16 [0.13, 0.18]	0.062 [0.060, 0.065]	4 of 10	4 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	0.75 [0.72, 0.79]	+1 [−40, +50]	83 [66, 104]	0.00 [−0.05, 0.04]	0.061 [0.058, 0.063]	2 of 10	3 of 10	0 of 10
Transformer	no_ar + kmeans_8	0.92 [0.77, 1.10]	+117 [+39, +206]	148 [88, 220]	0.12 [0.09, 0.16]	0.065 [0.061, 0.068]	2 of 10	2 of 10	0 of 10

Rate 0.01, churn 60% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	0.57 [0.54, 0.60]	+41 [+32, +50]	88 [83, 93]	0.26 [0.23, 0.30]	—	—	—	—
LSTM	no_ar	0.83 [0.79, 0.87]	+290 [+267, +316]	291 [269, 317]	0.01 [−0.03, 0.04]	0.052 [0.049, 0.055]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded	3.39 [1.42, 5.65]	+514 [+364, +705]	514 [365, 705]	−0.20 [−0.25, −0.15]	0.052 [0.049, 0.056]	0 of 10	0 of 10	0 of 10
LSTM	ar_bounded	0.71 [0.66, 0.76]	+190 [+150, +235]	196 [159, 240]	−0.04 [−0.10, 0.03]	0.051 [0.049, 0.054]	0 of 10	0 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	0.79 [0.74, 0.83]	+241 [+216, +266]	245 [222, 269]	0.07 [0.04, 0.11]	0.051 [0.048, 0.054]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	5.48 [1.95, 9.60]	+900 [+424, +1465]	904 [431, 1467]	−0.04 [−0.13, 0.04]	0.051 [0.048, 0.054]	0 of 10	0 of 10	0 of 10
LSTM	no_ar + kmeans_8	0.82 [0.76, 0.87]	+271 [+240, +305]	273 [243, 306]	0.16 [0.12, 0.20]	0.051 [0.048, 0.054]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	0.63 [0.60, 0.66]	+96 [+47, +149]	129 [97, 167]	0.07 [0.04, 0.09]	0.050 [0.047, 0.053]	3 of 10	3 of 10	0 of 10
Transformer	ar_unbounded	0.75 [0.62, 0.89]	<u>+68</u> [<u>+18</u> , <u>+124</u>]	127 [89, 169]	0.06 [0.02, 0.09]	0.049 [0.046, 0.052]	5 of 10	4 of 10	0 of 10
Transformer	ar_bounded	0.60 [0.57, 0.63]	<u>+50</u> [<u>+17</u> , <u>+83</u>]	<u>98</u> [<u>82</u> , <u>116</u>]	0.06 [0.02, 0.10]	0.049 [0.046, 0.053]	4 of 10	4 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	0.63 [0.58, 0.69]	<u>+43</u> [<u>−2</u> , <u>+91</u>]	<u>103</u> [<u>81</u> , <u>129</u>]	0.18 [0.12, 0.23]	0.047 [0.044, 0.050]	5 of 10	5 of 10	1 of 10
Transformer	ar_unbounded + kmeans_8	0.68 [0.60, 0.78]	+68 [+17, +130]	120 [92, 163]	0.10 [0.05, 0.14]	0.044 [0.042, 0.048]	3 of 10	2 of 10	0 of 10
Transformer	no_ar + kmeans_8	0.64 [0.60, 0.69]	+65 [+11, +119]	120 [93, 150]	0.16 [0.12, 0.21]	0.047 [0.044, 0.051]	4 of 10	2 of 10	0 of 10

Rate 0.01, churn 80% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	0.42 [0.38, 0.46]	+54 [+35, +76]	159 [137, 181]	0.20 [0.16, 0.24]	—	—	—	—
LSTM	no_ar	0.86 [0.79, 0.92]	+755 [+533, +1003]	759 [540, 1005]	−0.02 [−0.05, 0.02]	0.034 [0.031, 0.037]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded	1.53 [0.84, 2.57]	+821 [+501, +1141]	838 [533, 1147]	−0.12 [−0.15, −0.08]	0.034 [0.031, 0.037]	0 of 10	0 of 10	0 of 10
LSTM	ar_bounded	0.71 [0.62, 0.80]	+576 [+348, +831]	591 [374, 838]	−0.07 [−0.10, −0.04]	0.034 [0.031, 0.037]	0 of 10	0 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	0.63 [0.54, 0.72]	+429 [+304, +587]	445 [328, 598]	0.07 [0.02, 0.12]	0.033 [0.030, 0.036]	0 of 10	0 of 10	1 of 10
LSTM	ar_unbounded + kmeans_8	6.26 [1.38, 11.96]	+1993 [+731, +3553]	2015 [763, 3571]	-0.02 [-0.08, 0.04]	0.033 [0.030, 0.035]	0 of 10	0 of 10	0 of 10
LSTM	no_ar + kmeans_8	0.85 [0.79, 0.91]	+725 [+542, +937]	729 [548, 938]	0.11 [0.07, 0.16]	0.033 [0.030, 0.036]	0 of 10	0 of 10	1 of 10
Transformer	no_ar	<u>0.44</u> <u>[0.39,</u> <u>0.49]</u>	+121 [+66, +182]	197 [159, 239]	0.07 [0.02, 0.12]	0.031 [0.029, 0.034]	2 of 10	3 of 10	0 of 10
Transformer	ar_unbounded	0.70 [0.44, 1.18]	+264 [+73, +541]	345 [170, 611]	0.00 [-0.09, 0.08]	0.031 [0.028, 0.033]	3 of 10	3 of 10	0 of 10
Transformer	ar_bounded	<u>0.44</u> <u>[0.38,</u> <u>0.49]</u>	<u>+89</u> [<u>+45,</u> <u>+135]</u>	<u>176</u> <u>[152,</u> <u>201]</u>	0.08 [0.02, 0.14]	0.031 [0.028, 0.034]	4 of 10	4 of 10	1 of 10
Transformer	ar_bounded + kmeans_8	0.51 [0.43, 0.58]	+109 [+53, +183]	196 [159, 246]	0.11 [0.05, 0.16]	0.028 [0.026, 0.031]	4 of 10	4 of 10	2 of 10
Transformer	ar_unbounded + kmeans_8	0.51 [0.44, 0.59]	+120 [+28, +218]	216 [153, 283]	0.09 [0.03, 0.14]	0.027 [0.025, 0.030]	3 of 10	2 of 10	1 of 10
Transformer	no_ar + kmeans_8	<u>0.48</u> <u>[0.42,</u> <u>0.54]</u>	<u>+104</u> <u>[+32,</u> <u>+205]</u>	<u>195</u> <u>[144,</u> <u>271]</u>	0.15 [0.12, 0.18]	0.030 [0.027, 0.033]	6 of 10	5 of 10	2 of 10

Rate 0.05, churn 20% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	1.92 [1.89, 1.96]	+8 [+6, +11]	35 [34, 37]	0.58 [0.56, 0.60]	—	—	—	—
LSTM	no_ar	2.33 [2.20, 2.46]	+42 [+33, +51]	45 [37, 53]	0.29 [0.13, 0.44]	0.175 [0.173, 0.178]	2 of 10	0 of 10	0 of 10
LSTM	ar_unbounded	14.23 [11.69, 17.09]	+240 [+190, +299]	240 [190, 299]	0.44 [0.42, 0.46]	0.177 [0.175, 0.179]	0 of 10	0 of 10	0 of 10
LSTM	ar_bounded	2.29 [2.23, 2.33]	+47 [+39, +54]	48 [42, 55]	0.50 [0.47, 0.53]	0.174 [0.172, 0.176]	1 of 10	0 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	2.20 [2.13, 2.28]	+29 [+24, +33]	39 [37, 42]	0.45 [0.43, 0.47]	<u>0.169</u> [0.166, 0.172]	2 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	11.36 [9.16, 13.38]	+188 [+151, +224]	189 [151, 224]	0.46 [0.43, 0.49]	0.174 [0.171, 0.177]	0 of 10	0 of 10	0 of 10
LSTM	no_ar + kmeans_8	2.26 [2.17, 2.36]	+40 [+33, +48]	47 [41, 53]	0.46 [0.43, 0.50]	0.169 [0.167, 0.172]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	2.70 [2.53, 2.88]	+72 [+55, +87]	75 [62, 88]	0.31 [0.20, 0.40]	0.175 [0.173, 0.177]	1 of 10	1 of 10	0 of 10
Transformer	ar_unbounded	2.69 [2.46, 2.89]	+90 [+68, +109]	91 [70, 109]	0.52 [0.50, 0.54]	0.173 [0.171, 0.176]	1 of 10	0 of 10	0 of 10
Transformer	ar_bounded	2.56 [2.40, 2.69]	+61 [+38, +80]	68 [54, 80]	0.45 [0.42, 0.48]	0.174 [0.172, 0.176]	2 of 10	1 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	2.65 [2.50, 2.80]	+77 [+63, +89]	79 [67, 90]	0.51 [0.48, 0.53]	0.168 [0.165, 0.170]	0 of 10	0 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	2.66 [2.40, 2.91]	+61 [+36, +81]	69 [55, 82]	0.47 [0.40, 0.52]	<u>0.168</u> [0.166, 0.171]	2 of 10	1 of 10	0 of 10
Transformer	no_ar + kmeans_8	2.44 [2.25, 2.61]	+62 [+42, +81]	67 [51, 83]	0.49 [0.45, 0.52]	0.169 [0.166, 0.172]	2 of 10	0 of 10	0 of 10

Rate 0.05, churn 40% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	1.63 [1.60, 1.65]	+15 [+11, +19]	39 [37, 41]	0.60 [0.58, 0.61]	—	—	—	—
LSTM	no_ar	2.24 [2.15, 2.33]	+98 [+87, +109]	99 [88, 110]	0.20 [0.05, 0.34]	0.144 [0.142, 0.147]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded	10.38 [7.99, 12.92]	+313 [+255, +372]	313 [255, 372]	0.39 [0.36, 0.42]	0.145 [0.142, 0.147]	0 of 10	0 of 10	0 of 10
LSTM	ar_bounded	1.88 [1.82, 1.92]	+52 [+43, +61]	56 [47, 63]	0.54 [0.51, 0.56]	0.138 [0.136, 0.141]	1 of 10	0 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	2.00 [1.93, 2.09]	+61 [+50, +75]	64 [54, 78]	0.45 [0.43, 0.48]	0.135 [0.132, 0.138]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	9.40 [4.97, 14.54]	+221 [+101, +363]	229 [116, 367]	0.37 [0.32, 0.43]	0.140 [0.137, 0.143]	3 of 10	2 of 10	0 of 10
LSTM	no_ar + kmeans_8	2.09 [2.05, 2.14]	+81 [+71, +91]	82 [73, 92]	0.46 [0.42, 0.50]	0.137 [0.134, 0.139]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	2.28 [2.13, 2.44]	+108 [+83, +132]	110 [85, 132]	0.42 [0.38, 0.45]	0.142 [0.139, 0.144]	0 of 10	0 of 10	0 of 10
Transformer	ar_unbounded	2.30 [1.97, 2.75]	+101 [+72, +128]	103 [77, 129]	0.51 [0.44, 0.56]	0.138 [0.136, 0.141]	1 of 10	0 of 10	0 of 10
Transformer	ar_bounded	1.91 [1.74, 2.12]	+53 [+22, +88]	65 [40, 96]	0.55 [0.52, 0.57]	0.138 [0.136, 0.141]	3 of 10	2 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	2.00 [1.91, 2.10]	+48 [+32, +63]	56 [44, 68]	0.51 [0.49, 0.54]	0.133 [0.131, 0.136]	3 of 10	0 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	2.15 [1.98, 2.34]	+68 [+36, +101]	78 [54, 107]	0.46 [0.38, 0.52]	<u>0.134</u> <u>[0.131,</u> <u>0.137]</u>	2 of 10	3 of 10	0 of 10
Transformer	no_ar + kmeans_8	2.37 [2.20, 2.56]	+105 [+79, +134]	107 [82, 135]	0.48 [0.44, 0.51]	<u>0.134</u> <u>[0.131,</u> <u>0.137]</u>	0 of 10	0 of 10	0 of 10

Rate 0.05, churn 60% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	1.32 [1.27, 1.37]	+9 [+1, +16]	42 [40, 44]	0.56 [0.55, 0.57]	—	—	—	—
LSTM	no_ar	1.95 [1.86, 2.04]	+157 [+138, +176]	158 [139, 176]	0.03 [0.00, 0.06]	0.109 [0.106, 0.112]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded	10.76 [8.38, 12.94]	+462 [+338, +630]	462 [338, 630]	0.35 [0.32, 0.38]	0.108 [0.105, 0.111]	0 of 10	0 of 10	0 of 10
LSTM	ar_bounded	1.43 [1.37, 1.51]	+60 [+43, +74]	67 [57, 77]	0.52 [0.51, 0.53]	0.100 [0.098, 0.103]	1 of 10	1 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	1.80 [1.70, 1.90]	+108 [+88, +129]	110 [92, 131]	0.43 [0.39, 0.46]	0.098 [0.095, 0.101]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	8.50 [5.06, 12.13]	+357 [+173, +578]	359 [176, 580]	0.36 [0.28, 0.43]	0.104 [0.101, 0.107]	0 of 10	0 of 10	0 of 10
LSTM	no_ar + kmeans_8	1.99 [1.89, 2.10]	+172 [+153, +191]	172 [153, 191]	0.38 [0.31, 0.43]	0.102 [0.099, 0.104]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	1.87 [1.69, 2.04]	+144 [+89, +197]	149 [98, 198]	0.37 [0.32, 0.42]	0.105 [0.102, 0.107]	1 of 10	1 of 10	0 of 10
Transformer	ar_unbounded	1.73 [1.61, 1.85]	+121 [+87, +156]	124 [92, 157]	0.49 [0.44, 0.52]	0.100 [0.098, 0.103]	0 of 10	0 of 10	0 of 10
Transformer	ar_bounded	1.47 [1.39, 1.55]	+66 [+36, +96]	77 [56, 101]	0.52 [0.50, 0.53]	0.100 [0.097, 0.103]	2 of 10	0 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	2.02 [1.81, 2.21]	+152 [+108, +196]	155 [114, 197]	0.47 [0.44, 0.50]	<u>0.096</u> <u>[0.093,</u> <u>0.098]</u>	0 of 10	0 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	1.90 [1.64, 2.31]	+98 [+55, +159]	103 [63, 163]	0.43 [0.36, 0.49]	0.094 [0.092, 0.097]	0 of 10	0 of 10	0 of 10
Transformer	no_ar + kmeans_8	1.91 [1.73, 2.08]	+131 [+88, +177]	134 [96, 178]	0.48 [0.46, 0.50]	0.096 [0.093, 0.098]	0 of 10	0 of 10	0 of 10

Rate 0.05, churn 80% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	0.89 [0.86, 0.92]	+13 [0, +26]	58 [53, 64]	0.44 [0.42, 0.47]	—	—	—	—
LSTM	no_ar	1.88 [1.75, 2.03]	+497 [+451, +544]	497 [451, 544]	−0.01 [−0.03, 0.02]	0.070 [0.065, 0.074]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded	12.14 [7.39, 16.74]	+1314 [+789, +1810]	1315 [789, 1810]	0.26 [0.20, 0.33]	0.067 [0.062, 0.072]	0 of 10	0 of 10	0 of 10
LSTM	ar_bounded	0.97 [0.93, 1.01]	<u>+19 [+</u> <u>1,</u> <u>+37]</u>	<u>60 [53,</u> <u>69]</u>	0.40 [0.37, 0.43]	0.061 [0.057, 0.065]	7 of 10	4 of 10	1 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	1.58 [1.36, 1.82]	+287 [+189, +396]	290 [195, 397]	0.36 [0.34, 0.38]	0.061 [0.057, 0.064]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	11.44 [7.94, 14.72]	+908 [+568, +1222]	919 [594, 1225]	0.32 [0.27, 0.37]	0.065 [0.061, 0.069]	0 of 10	0 of 10	0 of 10
LSTM	no_ar + kmeans_8	1.99 [1.90, 2.09]	+506 [+437, +560]	506 [437, 560]	0.34 [0.30, 0.38]	0.064 [0.060, 0.068]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	1.13 [1.07, 1.18]	+110 [+66, +157]	122 [87, 162]	0.32 [0.27, 0.35]	0.066 [0.062, 0.070]	2 of 10	2 of 10	0 of 10
Transformer	ar_unbounded	1.09 [1.00, 1.21]	+70 [+27, +122]	100 [69, 142]	0.19 [0.05, 0.34]	0.062 [0.058, 0.066]	1 of 10	1 of 10	0 of 10
Transformer	ar_bounded	1.04 [0.99, 1.08]	+76 [+28, +127]	102 [68, 141]	0.40 [0.38, 0.44]	0.061 [0.057, 0.065]	2 of 10	1 of 10	1 of 10
Transformer	ar_bounded + kmeans_8	1.11 [1.06, 1.14]	+62 [+15, +111]	94 [64, 128]	0.37 [0.34, 0.41]	0.058 [0.054, 0.061]	3 of 10	3 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	1.15 [0.98, 1.43]	+63 [+14, +129]	97 [63, 151]	0.38 [0.36, 0.41]	0.057 [0.053, 0.061]	3 of 10	3 of 10	0 of 10
Transformer	no_ar + kmeans_8	1.29 [1.12, 1.47]	+161 [+69, +254]	181 [103, 262]	0.36 [0.30, 0.40]	0.059 [0.055, 0.063]	1 of 10	2 of 10	0 of 10

Rate 0.10, churn 20% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	2.91 [2.86, 2.95]	+3 [+1, +4]	31 [30, 32]	0.72 [0.71, 0.73]	—	—	—	—
LSTM	no_ar	3.28 [3.16, 3.41]	+31 [+24, +38]	<u>33</u> [26, 40]	0.67 [0.65, 0.69]	0.262 [0.255, 0.268]	4 of 10	0 of 10	0 of 10
LSTM	ar_unbounded	21.15 [15.92, 26.87]	+301 [+223, +391]	301 [223, 391]	0.60 [0.58, 0.62]	0.272 [0.265, 0.278]	0 of 10	0 of 10	0 of 10
LSTM	ar_bounded	3.32 [3.21, 3.45]	+29 [+17, +39]	<u>32</u> [24, 41]	0.69 [0.68, 0.70]	0.261 [0.255, 0.266]	4 of 10	0 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	3.46 [3.40, 3.53]	+28 [+24, +33]	<u>33</u> [29, <u>37</u>]	0.60 [0.57, 0.64]	0.254 [0.247, 0.260]	4 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	13.08 [8.43, 17.91]	+141 [+82, +205]	145 [88, 206]	0.58 [0.54, 0.61]	0.265 [0.258, 0.270]	0 of 10	0 of 10	0 of 10
LSTM	no_ar + kmeans_8	3.57 [3.46, 3.66]	+35 [+28, +42]	39 [34, 44]	0.58 [0.55, 0.61]	0.258 [0.251, 0.265]	2 of 10	0 of 10	0 of 10
Transformer	no_ar	4.11 [3.84, 4.34]	+54 [+35, +72]	58 [45, 72]	0.58 [0.56, 0.60]	0.266 [0.259, 0.273]	2 of 10	0 of 10	0 of 10
Transformer	ar_unbounded	5.00 [4.30, 5.90]	+101 [+76, +128]	102 [77, 129]	0.65 [0.63, 0.66]	0.262 [0.255, 0.268]	0 of 10	0 of 10	0 of 10
Transformer	ar_bounded	3.74 [3.56, 3.95]	+43 [+23, +64]	52 [37, 68]	0.68 [0.67, 0.69]	0.262 [0.255, 0.268]	4 of 10	0 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	3.71 [3.46, 3.97]	+36 [+20, +54]	45 [35, 58]	0.65 [0.59, 0.68]	<u>0.255</u> <u>[0.247,</u> <u>0.260]</u>	2 of 10	1 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	3.86 [3.59, 4.13]	+38 [+22, +52]	47 [38, 55]	0.63 [0.57, 0.67]	<u>0.256</u> <u>[0.248,</u> <u>0.263]</u>	2 of 10	0 of 10	0 of 10
Transformer	no_ar + kmeans_8	3.98 [3.69, 4.30]	+54 [+38, +70]	57 [44, 71]	0.61 [0.57, 0.64]	0.256 [0.249, 0.261]	1 of 10	0 of 10	0 of 10

Rate 0.10, churn 40% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	2.47 [2.39, 2.55]	+1 [-2, +5]	32 [30, 33]	0.73 [0.72, 0.74]	—	—	—	—
LSTM	no_ar	3.11 [2.77, 3.61]	+57 [+36, +88]	58 [38, 89]	0.64 [0.57, 0.69]	0.205 [0.200, 0.209]	2 of 10	0 of 10	0 of 10
LSTM	ar_unbounded	18.03 [13.16, 23.69]	+375 [+302, +471]	375 [302, 471]	0.57 [0.56, 0.59]	0.213 [0.210, 0.217]	0 of 10	0 of 10	0 of 10
LSTM	ar_bounded	2.79 [2.68, 2.89]	+28 [+18, +39]	<u>34</u> [27, <u>42]</u>	0.69 [0.68, 0.71]	0.198 [0.194, 0.202]	6 of 10	1 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	3.15 [3.03, 3.32]	+52 [+45, +58]	53 [48, 59]	0.64 [0.62, 0.65]	<u>0.193</u> [0.190, 0.197]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	9.72 [5.58, 14.13]	+167 [+89, +250]	174 [101, 253]	0.56 [0.52, 0.61]	0.203 [0.198, 0.207]	0 of 10	0 of 10	0 of 10
LSTM	no_ar + kmeans_8	3.42 [3.33, 3.53]	+74 [+68, +80]	74 [69, 80]	0.57 [0.53, 0.61]	0.198 [0.194, 0.202]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	3.69 [3.45, 3.93]	+86 [+51, +117]	94 [67, 119]	0.58 [0.57, 0.59]	0.206 [0.202, 0.210]	1 of 10	0 of 10	0 of 10
Transformer	ar_unbounded	3.89 [3.30, 4.61]	+89 [+67, +115]	91 [69, 116]	0.64 [0.58, 0.67]	0.200 [0.196, 0.205]	0 of 10	0 of 10	0 of 10
Transformer	ar_bounded	2.87 [2.64, 3.15]	+27 [+9, +43]	41 [32, 50]	0.64 [0.54, 0.70]	0.199 [0.195, 0.203]	4 of 10	0 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	3.42 [3.12, 3.76]	+51 [+31, +72]	58 [43, 74]	0.65 [0.64, 0.66]	0.192 [0.188, 0.196]	1 of 10	0 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	3.74 [3.29, 4.24]	+72 [+47, +99]	77 [55, 100]	0.62 [0.58, 0.65]	<u>0.192</u> [0.188, 0.197]	1 of 10	0 of 10	0 of 10
Transformer	no_ar + kmeans_8	3.99 [3.64, 4.28]	+93 [+74, +109]	95 [78, 109]	0.63 [0.61, 0.64]	0.195 [0.192, 0.199]	0 of 10	0 of 10	0 of 10

Rate 0.10, churn 60% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	2.00 [1.95, 2.06]	+3 [-3, +10]	37 [35, 39]	0.65 [0.64, 0.66]	—	—	—	—
LSTM	no_ar	2.43 [2.28, 2.61]	+61 [+33, +90]	69 [47, 93]	0.60 [0.58, 0.61]	0.149 [0.143, 0.153]	3 of 10	2 of 10	0 of 10
LSTM	ar_unbounded	7.56 [4.76, 10.97]	+299 [+164, +429]	307 [179, 431]	0.52 [0.47, 0.57]	0.154 [0.146, 0.160]	1 of 10	1 of 10	0 of 10
LSTM	ar_bounded	2.12 [2.05, 2.18]	+20 [+13, +28]	33 [30, 36]	0.63 [0.61, 0.64]	0.140 [0.136, 0.143]	7 of 10	1 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	2.79 [2.68, 2.91]	+92 [+74, +111]	93 [77, 112]	0.53 [0.50, 0.56]	0.139 [0.135, 0.141]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	6.06 [3.69, 8.57]	+133 [+53, +234]	142 [66, 240]	0.55 [0.51, 0.59]	0.146 [0.142, 0.151]	1 of 10	0 of 10	0 of 10
LSTM	no_ar + kmeans_8	3.12 [2.98, 3.28]	+132 [+110, +161]	132 [110, 161]	0.51 [0.46, 0.54]	0.146 [0.142, 0.149]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	2.72 [2.49, 3.00]	+98 [+59, +137]	103 [70, 139]	0.52 [0.50, 0.55]	0.150 [0.146, 0.153]	1 of 10	1 of 10	0 of 10
Transformer	ar_unbounded	3.57 [3.04, 4.12]	+159 [+124, +198]	159 [125, 198]	0.57 [0.54, 0.60]	0.143 [0.139, 0.146]	0 of 10	0 of 10	0 of 10
Transformer	ar_bounded	2.74 [2.40, 3.18]	+104 [+55, +152]	115 [79, 156]	0.61 [0.59, 0.62]	0.140 [0.136, 0.143]	1 of 10	0 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	3.05 [2.76, 3.40]	+106 [+73, +142]	108 [77, 143]	0.57 [0.55, 0.59]	0.135 [0.132, 0.138]	0 of 10	0 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	2.83 [2.56, 3.09]	+70 [+43, +97]	76 [55, 101]	0.56 [0.52, 0.59]	<u>0.136</u> <u>[0.132,</u> <u>0.138]</u>	1 of 10	1 of 10	0 of 10
Transformer	no_ar + kmeans_8	3.01 [2.75, 3.27]	+106 [+78, +132]	109 [82, 133]	0.56 [0.55, 0.58]	0.139 [0.135, 0.141]	0 of 10	0 of 10	0 of 10

Rate 0.10, churn 80% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	1.37 [1.26, 1.47]	<u>+7 [-7,</u> <u>+23]</u>	49 [44, 57]	0.51 [0.49, 0.52]	—	—	—	—
LSTM	no_ar	2.02 [1.73, 2.31]	+224 [+111, +362]	233 [128, 367]	0.27 [0.16, 0.39]	0.091 [0.085, 0.095]	2 of 10	2 of 10	0 of 10
LSTM	ar_unbounded	4.09 [2.30, 6.63]	+285 [+103, +493]	307 [137, 503]	0.43 [0.38, 0.47]	0.085 [0.080, 0.089]	1 of 10	1 of 10	0 of 10
LSTM	ar_bounded	1.45 [1.31, 1.58]	+4 [-7, +15]	45 [40, 50]	0.47 [0.45, 0.49]	0.077 [0.073, 0.081]	9 of 10	8 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	2.04 [1.89, 2.16]	+131 [+97, +177]	134 [102, 178]	0.45 [0.42, 0.47]	<u>0.076</u> [<u>0.072</u> , <u>0.079</u>]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	3.98 [1.79, 6.39]	+146 [+2, +355]	189 [56, 382]	0.45 [0.42, 0.47]	0.078 [0.074, 0.084]	1 of 10	2 of 10	0 of 10
LSTM	no_ar + kmeans_8	2.42 [2.27, 2.58]	+309 [+231, +404]	309 [231, 404]	0.38 [0.34, 0.42]	0.086 [0.083, 0.089]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	2.20 [2.04, 2.36]	+265 [+201, +342]	265 [202, 343]	0.41 [0.39, 0.43]	0.088 [0.084, 0.091]	0 of 10	0 of 10	0 of 10
Transformer	ar_unbounded	2.70 [2.27, 3.19]	+245 [+199, +300]	245 [199, 300]	0.45 [0.42, 0.48]	0.080 [0.077, 0.084]	0 of 10	0 of 10	0 of 10
Transformer	ar_bounded	1.57 [1.46, 1.66]	+82 [+34, +135]	102 [68, 146]	0.47 [0.45, 0.48]	0.078 [0.075, 0.082]	3 of 10	2 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	2.31 [2.08, 2.56]	+175 [+130, +223]	177 [133, 223]	0.42 [0.38, 0.45]	0.075 [0.072, 0.078]	0 of 10	0 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	2.21 [1.85, 2.61]	+142 [+87, +200]	150 [103, 204]	0.44 [0.41, 0.46]	<u>0.075</u> [<u>0.072</u> , <u>0.079</u>]	1 of 10	0 of 10	0 of 10
Transformer	no_ar + kmeans_8	2.39 [2.16, 2.68]	+201 [+136, +275]	203 [140, 276]	0.43 [0.40, 0.45]	0.077 [0.073, 0.080]	0 of 10	0 of 10	0 of 10

Rate 0.30, churn 20% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	6.93 [6.72, 7.13]	−14 [−16, −13]	30 [30, 31]	0.86 [0.85, 0.87]	—	—	—	—
LSTM	no_ar	6.92 [6.65, 7.21]	+17 [+12, +22]	19 [16, 22]	0.85 [0.85, 0.86]	0.497 [0.491, 0.504]	10 of 10	2 of 10	0 of 10
LSTM	ar_unbounded	35.67 [30.70, 39.91]	+268 [+233, +302]	268 [233, 302]	0.73 [0.73, 0.74]	0.546 [0.539, 0.553]	0 of 10	0 of 10	0 of 10
LSTM	ar_bounded	6.49 [6.26, 6.75]	+10 [+7, +12]	12 [10, 13]	0.85 [0.85, 0.86]	0.493 [0.486, 0.499]	10 of 10	8 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	7.96 [7.66, 8.24]	+25 [+22, +28]	26 [23, 29]	0.79 [0.78, 0.80]	0.490 [0.483, 0.497]	9 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	15.60 [11.32, 21.66]	+76 [+46, +107]	80 [54, 109]	0.75 [0.73, 0.78]	0.513 [0.503, 0.523]	3 of 10	2 of 10	0 of 10
LSTM	no_ar + kmeans_8	8.31 [8.05, 8.55]	+27 [+25, +30]	28 [25, 31]	0.75 [0.73, 0.76]	0.493 [0.486, 0.500]	7 of 10	0 of 10	0 of 10
Transformer	no_ar	8.30 [7.55, 9.39]	+20 [+3, +40]	36 [26, 50]	0.71 [0.70, 0.73]	0.517 [0.511, 0.523]	8 of 10	5 of 10	0 of 10
Transformer	ar_unbounded	12.13 [9.75, 14.58]	+69 [+47, +93]	71 [51, 94]	0.76 [0.71, 0.80]	0.511 [0.504, 0.517]	0 of 10	0 of 10	0 of 10
Transformer	ar_bounded	8.06 [7.06, 9.18]	+25 [+9, +43]	37 [27, 49]	0.81 [0.79, 0.82]	0.501 [0.494, 0.508]	6 of 10	5 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	8.08 [7.66, 8.47]	+25 [+17, +32]	32 [29, 37]	0.76 [0.70, 0.79]	0.494 [0.488, 0.501]	4 of 10	2 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	8.84 [7.98, 9.67]	+30 [+19, +40]	37 [32, 43]	0.80 [0.79, 0.80]	0.499 [0.491, 0.507]	2 of 10	3 of 10	0 of 10
Transformer	no_ar + kmeans_8	8.20 [7.80, 8.75]	+21 [+14, +30]	30 [26, 36]	0.76 [0.75, 0.77]	0.505 [0.498, 0.512]	7 of 10	4 of 10	0 of 10

Rate 0.30, churn 40% (10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	5.99 [5.79, 6.20]	−14 [−15, −12]	31 [30, 32]	0.83 [0.82, 0.84]	—	—	—	—
LSTM	no_ar	<u>5.74</u> <u>[5.53,</u> <u>5.96]</u>	<u>+7</u> [<u>+3,</u> <u>+11]</u>	14 [13, 16]	0.81 [0.81, 0.82]	0.361 [0.356, 0.366]	10 of 10	8 of 10	0 of 10
LSTM	ar_unbounded	18.04 [11.28, 25.33]	+151 [+62, +243]	161 [81, 247]	0.74 [0.71, 0.76]	0.400 [0.387, 0.412]	1 of 10	0 of 10	0 of 10
LSTM	ar_bounded	5.66 [5.46, 5.87]	+2 [−3, +6]	13 [11, 14]	0.82 [0.81, 0.82]	0.355 [0.350, 0.361]	10 of 10	9 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	7.67 [7.32, 8.04]	+44 [+38, +49]	45 [39, 50]	0.75 [0.74, 0.77]	0.358 [0.353, 0.364]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	10.25 [7.51, 14.34]	+43 [+8, +84]	62 [36, 95]	0.75 [0.72, 0.77]	0.380 [0.373, 0.387]	3 of 10	4 of 10	0 of 10
LSTM	no_ar + kmeans_8	8.24 [7.80, 8.73]	+53 [+48, +58]	53 [48, 58]	0.74 [0.73, 0.76]	0.361 [0.356, 0.366]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	7.34 [6.87, 7.88]	+28 [+16, +41]	36 [28, 45]	0.68 [0.66, 0.70]	0.386 [0.380, 0.392]	4 of 10	2 of 10	0 of 10
Transformer	ar_unbounded	10.23 [8.64, 11.86]	+81 [+51, +108]	86 [62, 109]	0.69 [0.62, 0.75]	0.376 [0.370, 0.383]	1 of 10	1 of 10	0 of 10
Transformer	ar_bounded	6.88 [6.14, 7.73]	+23 [+2, +45]	42 [32, 55]	0.70 [0.60, 0.78]	0.364 [0.357, 0.369]	4 of 10	3 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	7.90 [7.56, 8.24]	+45 [+34, +55]	47 [39, 56]	0.74 [0.73, 0.76]	0.357 [0.351, 0.363]	1 of 10	1 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	7.85 [7.03, 8.83]	+21 [+5, +39]	41 [33, 52]	0.74 [0.70, 0.76]	0.363 [0.358, 0.368]	4 of 10	3 of 10	0 of 10
Transformer	no_ar + kmeans_8	8.52 [7.90, 9.09]	+53 [+42, +63]	55 [45, 63]	0.70 [0.69, 0.72]	0.373 [0.367, 0.378]	1 of 10	0 of 10	0 of 10

Rate 0.30, churn 60% (9–10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	4.72 [4.60, 4.86]	<u>-12</u> [-16, -8]	32 [31, 33]	0.77 [0.76, 0.77]	—	—	—	—
LSTM	no_ar	4.73 [4.53, 4.96]	-1 [-8, +5]	18 [15, 21]	0.70 [0.69, 0.71]	0.237 [0.231, 0.241]	10 of 10	7 of 10	0 of 10
LSTM	ar_unbounded	7.04 [6.23, 7.91]	+56 [+36, +77]	64 [49, 80]	0.66 [0.64, 0.67]	0.259 [0.251, 0.268]	1 of 10	1 of 10	0 of 10
LSTM	ar_bounded	4.60 [4.46, 4.77]	<u>-7</u> [-13, +1]	<u>19</u> [17, 22]	0.70 [0.69, 0.71]	0.232 [0.227, 0.237]	10 of 10	6 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	6.73 [6.43, 7.06]	+71 [+63, +78]	71 [64, 78]	0.66 [0.64, 0.67]	0.236 [0.230, 0.240]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	6.49 [5.73, 7.54]	+11 [−16, +44]	54 [41, 71]	0.66 [0.65, 0.67]	0.253 [0.245, 0.261]	1 of 10	2 of 10	0 of 10
LSTM	no_ar + kmeans_8	7.35 [6.96, 7.68]	+90 [+81, +99]	90 [81, 99]	0.64 [0.63, 0.65]	0.242 [0.235, 0.248]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	6.19 [5.78, 6.68]	+40 [+11, +68]	57 [41, 75]	0.59 [0.57, 0.60]	0.265 [0.258, 0.270]	3 of 10	2 of 10	0 of 10
Transformer	ar_unbounded	7.95 [7.26, 8.74]	+96 [+72, +121]	97 [73, 122]	0.63 [0.59, 0.66]	0.251 [0.242, 0.259]	0 of 9	0 of 9	0 of 9
Transformer	ar_bounded	5.81 [5.27, 6.37]	+59 [+31, +88]	67 [46, 91]	0.67 [0.65, 0.68]	0.238 [0.232, 0.243]	3 of 10	1 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	7.33 [6.46, 8.41]	+83 [+61, +108]	85 [64, 109]	0.64 [0.62, 0.66]	0.234 [0.228, 0.239]	0 of 10	0 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	7.05 [6.19, 8.11]	+54 [+27, +82]	65 [47, 88]	0.64 [0.62, 0.66]	0.241 [0.235, 0.247]	1 of 10	3 of 10	0 of 10
Transformer	no_ar + kmeans_8	6.74 [6.36, 7.15]	+65 [+52, +80]	65 [53, 80]	0.61 [0.60, 0.62]	0.249 [0.243, 0.255]	0 of 10	0 of 10	0 of 10

Rate 0.30, churn 80% (9–10 panels)

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
Pareto/NBD	—	<u>3.03</u> [2.86, 3.22]	−12 [−18, −5]	<u>35</u> [34, 36]	0.70 [0.68, 0.72]	—	—	—	—
LSTM	no_ar	3.26 [3.04, 3.49]	<u>−15</u> [−22, −8]	32 [28, 36]	0.51 [0.49, 0.53]	0.121 [0.116, 0.127]	7 of 10	3 of 10	0 of 10
LSTM	ar_unbounded	4.63 [3.48, 6.06]	+50 [+12, +95]	70 [41, 107]	0.48 [0.46, 0.50]	0.131 [0.126, 0.137]	5 of 10	4 of 10	0 of 10
LSTM	ar_bounded	3.01 [2.85, 3.19]	<u>−13</u> [−23, −1]	35 [33, 39]	0.51 [0.48, 0.53]	0.117 [0.110, 0.123]	5 of 10	4 of 10	0 of 10

Model	Arm	RMSE	Bias %	MAPE	Spearman	Val. CE	Beats P/NBD: MAPE	Beats P/NBD: bias	Beats P/NBD: Spearman
LSTM	ar_bounded + kmeans_8	4.08 [3.89, 4.26]	+83 [+67, +100]	84 [69, 101]	0.49 [0.46, 0.51]	0.118 [0.111, 0.125]	0 of 10	0 of 10	0 of 10
LSTM	ar_unbounded + kmeans_8	3.89 [3.52, 4.25]	0 [-17, +18]	47 [41, 53]	0.48 [0.47, 0.50]	0.126 [0.119, 0.133]	1 of 10	3 of 10	0 of 10
LSTM	no_ar + kmeans_8	5.34 [4.83, 5.86]	+142 [+115, +170]	142 [115, 170]	0.48 [0.46, 0.49]	0.127 [0.119, 0.134]	0 of 10	0 of 10	0 of 10
Transformer	no_ar	4.06 [3.69, 4.49]	+86 [+43, +133]	96 [62, 138]	0.44 [0.42, 0.46]	0.141 [0.133, 0.148]	2 of 10	2 of 10	0 of 10
Transformer	ar_unbounded	4.84 [3.90, 6.23]	+112 [+61, +168]	118 [71, 171]	0.43 [0.40, 0.46]	0.128 [0.120, 0.136]	1 of 9	1 of 9	0 of 9
Transformer	ar_bounded	3.18 [2.98, 3.38]	+46 [+30, +61]	54 [45, 64]	0.48 [0.46, 0.50]	0.119 [0.112, 0.125]	2 of 10	2 of 10	0 of 10
Transformer	ar_bounded + kmeans_8	4.54 [3.83, 5.33]	+87 [+56, +122]	91 [61, 123]	0.47 [0.44, 0.50]	0.118 [0.112, 0.124]	1 of 10	1 of 10	0 of 10
Transformer	ar_unbounded + kmeans_8	4.26 [3.89, 4.65]	+39 [+19, +57]	56 [48, 65]	0.47 [0.44, 0.49]	0.123 [0.115, 0.130]	0 of 10	1 of 10	0 of 10
Transformer	no_ar + kmeans_8	4.96 [4.36, 5.63]	+119 [+88, +148]	120 [90, 148]	0.45 [0.44, 0.47]	0.129 [0.122, 0.137]	0 of 10	0 of 10	0 of 10

The three best trees per cell, by MAPE

Ranked by mean MAPE over the cell's 10 panels (lowest first); MAPE carries its 95% bootstrap interval, the other metrics are means. "Δ vs rank 1" is the paired difference of that tree from the cell's leader on the same panels, with its 95% bootstrap interval; "clearly behind" is yes when that interval excludes 0. An interval end printed as 0 is not exactly 0 before rounding. The "Beats P/NBD" counts are over the cell's panels.

Rate	Churn	Rank	Model	Arm	MAPE	Spearman	RMSE	Bias %	Val. CE	Δ vs rank 1 [95% CI]	Clearly behind rank 1	Beats P/NBD: MAPE / bias / Spearman
0.01	20%	1	Pareto/NBD	—	48 [45, 51]	0.22	0.79	+12	—	—	—	—
		2	Transformer	ar_unbounded	49 [44, 56]	−0.05	0.83	+7	0.077	+1 [−3, +7]	no	6 / 6 / 0 of 10

Rate	Churn	Rank	Model	Arm	MAPE	Spearman	RMSE	Bias %	Val. CE	Δ vs rank 1 [95% CI]	Clearly behind rank 1	Beats P/NBD: MAPE / bias / Spearman
		3	Transformer	no_ar + kmeans_8	55 [49, 63]	0.12	0.86	-3	0.077	+7 [+1, +15]	yes	4 / 2 / 0 of 10
0.01	40%	1	Pareto/NBD	—	66 [62, 68]	0.25	0.68	+34	—	—	—	—
		2	Transformer	ar_bounded	75 [65, 87]	-0.02	0.71	+27	0.065	+10 [0, +20]	no	4 / 2 / 0 of 10
		3	Transformer	ar_unbounded	78 [59, 100]	0.00	0.72	+18	0.065	+12 [-5, +33]	no	6 / 5 / 1 of 10
0.01	60%	1	Pareto/NBD	—	88 [83, 93]	0.26	0.57	+41	—	—	—	—
		2	Transformer	ar_bounded	98 [82, 116]	0.06	0.60	+50	0.049	+10 [-7, +29]	no	4 / 4 / 0 of 10
		3	Transformer	ar_bounded + kmeans_8	103 [81, 129]	0.18	0.63	+43	0.047	+15 [-7, +41]	no	5 / 5 / 1 of 10
0.01	80%	1	Pareto/NBD	—	159 [137, 181]	0.20	0.42	+54	—	—	—	—
		2	Transformer	ar_bounded	176 [152, 201]	0.08	0.44	+89	0.031	+17 [-12, +49]	no	4 / 4 / 1 of 10
		3	Transformer	no_ar + kmeans_8	195 [144, 271]	0.15	0.48	+104	0.030	+36 [-8, +103]	no	6 / 5 / 2 of 10
0.05	20%	1	Pareto/NBD	—	35 [34, 37]	0.58	1.92	+8	—	—	—	—
		2	LSTM	ar_bounded + kmeans_8	39 [37, 42]	0.45	2.20	+29	0.169	+4 [+2, +6]	yes	2 / 0 / 0 of 10
		3	LSTM	no_ar	45 [37, 53]	0.29	2.33	+42	0.175	+9 [+2, +17]	yes	2 / 0 / 0 of 10
0.05	40%	1	Pareto/NBD	—	39 [37, 41]	0.60	1.63	+15	—	—	—	—

Rate	Churn	Rank	Model	Arm	MAPE	Spearman	RMSE	Bias %	Val. CE	Δ vs rank 1 [95% CI]	Clearly behind rank 1	Beats P/NBD: MAPE / bias / Spearman
		2	LSTM	ar_bounded	56 [47, 63]	0.54	1.88	+52	0.138	+17 [+9, +24]	yes	1 / 0 / 0 of 10
		3	Transformer	ar_bounded + kmeans_8	56 [44, 68]	0.51	2.00	+48	0.133	+17 [+6, +29]	yes	3 / 0 / 0 of 10
0.05	60%	1	Pareto/NBD	—	42 [40, 44]	0.56	1.32	+9	—	—	—	—
		2	LSTM	ar_bounded	67 [57, 77]	0.52	1.43	+60	0.100	+24 [+15, +34]	yes	1 / 1 / 0 of 10
		3	Transformer	ar_bounded	77 [56, 101]	0.52	1.47	+66	0.100	+35 [+15, +58]	yes	2 / 0 / 0 of 10
0.05	80%	1	Pareto/NBD	—	58 [53, 64]	0.44	0.89	+13	—	—	—	—
		2	LSTM	ar_bounded	60 [53, 69]	0.40	0.97	+19	0.061	+2 [-2, +8]	no	7 / 4 / 1 of 10
		3	Transformer	ar_bounded + kmeans_8	94 [64, 128]	0.37	1.11	+62	0.058	+36 [+9, +67]	yes	3 / 3 / 0 of 10
0.10	20%	1	Pareto/NBD	—	31 [30, 32]	0.72	2.91	+3	—	—	—	—
		2	LSTM	ar_bounded	32 [24, 41]	0.69	3.32	+29	0.261	+1 [-7, +9]	no	4 / 0 / 0 of 10
		3	LSTM	ar_bounded + kmeans_8	33 [29, 37]	0.60	3.46	+28	0.254	+2 [-2, +5]	no	4 / 0 / 0 of 10
0.10	40%	1	Pareto/NBD	—	32 [30, 33]	0.73	2.47	+1	—	—	—	—
		2	LSTM	ar_bounded	34 [27, 42]	0.69	2.79	+28	0.198	+3 [-5, +11]	no	6 / 1 / 0 of 10
		3	Transformer	ar_bounded	41 [32, 50]	0.64	2.87	+27	0.199	+9 [+1, +17]	yes	4 / 0 / 0 of 10

Rate	Churn	Rank	Model	Arm	MAPE	Spearman	RMSE	Bias %	Val. CE	Δ vs rank 1 [95% CI]	Clearly behind rank 1	Beats P/NBD: MAPE / bias / Spearman
0.10	60%	1	LSTM	ar_bounded	33 [30, 36]	0.63	2.12	+20	0.140	—	—	7 / 1 / 0 of 10
		2	Pareto/NBD	—	37 [35, 39]	0.65	2.00	+3	—	+4 [+1, +7]	yes	—
		3	LSTM	no_ar	69 [47, 93]	0.60	2.43	+61	0.149	+36 [+13, +62]	yes	3 / 2 / 0 of 10
0.10	80%	1	LSTM	ar_bounded	45 [40, 50]	0.47	1.45	+4	0.077	—	—	9 / 8 / 0 of 10
		2	Pareto/NBD	—	49 [44, 57]	0.51	1.37	+7	—	+4 [+1, +8]	yes	—
		3	Transformer	ar_bounded	102 [68, 146]	0.47	1.57	+82	0.078	+57 [+24, +97]	yes	3 / 2 / 0 of 10
0.30	20%	1	LSTM	ar_bounded	12 [10, 13]	0.85	6.49	+10	0.493	—	—	10 / 8 / 0 of 10
		2	LSTM	no_ar	19 [16, 22]	0.85	6.92	+17	0.497	+8 [+3, +12]	yes	10 / 2 / 0 of 10
		3	LSTM	ar_bounded + kmeans_8	26 [23, 29]	0.79	7.96	+25	0.490	+14 [+10, +17]	yes	9 / 0 / 0 of 10
0.30	40%	1	LSTM	ar_bounded	13 [11, 14]	0.82	5.66	+2	0.355	—	—	10 / 9 / 0 of 10
		2	LSTM	no_ar	14 [13, 16]	0.81	5.74	+7	0.361	+2 [0, +3]	yes	10 / 8 / 0 of 10
		3	Pareto/NBD	—	31 [30, 32]	0.83	5.99	−14	—	+18 [+17, +20]	yes	—
0.30	60%	1	LSTM	no_ar	18 [15, 21]	0.70	4.73	−1	0.237	—	—	10 / 7 / 0 of 10
		2	LSTM	ar_bounded	19 [17, 22]	0.70	4.60	−7	0.232	+1 [−1, +4]	no	10 / 6 / 0 of 10
		3	Pareto/NBD	—	32 [31,	0.77	4.72	−12	—	+14 [+11,	yes	—

Rate	Churn	Rank	Model	Arm	MAPE	Spearman	RMSE	Bias %	Val. CE	Δ vs rank 1 [95% CI]	Clearly behind rank 1	Beats P/NBD: MAPE / bias / Spearman
					33]					+16]		
0.30	80%	1	LSTM	no_ar	32 [28, 36]	0.51	3.26	-15	0.121	—	—	7 / 3 / 0 of 10
		2	Pareto/NBD	—	35 [34, 36]	0.70	3.03	-12	—	+3 [0, +6]	no	—
		3	LSTM	ar_bounded	35 [33, 39]	0.51	3.01	-13	0.117	+3 [+1, +6]	yes	5 / 4 / 0 of 10

The three best trees per cell, by Spearman

Ranked by mean per-customer Spearman (highest first); layout as above.

Rate	Churn	Rank	Model	Arm	MAPE	Spearman	RMSE	Bias %	Val. CE	Δ vs rank 1 [95% CI]	Clearly behind rank 1	Beats P/NBD: MAPE / bias / Spearman
0.01	20%	1	Pareto/NBD	—	48	0.22 [0.19, 0.25]	0.79	+12	—	—	—	—
		2	Transformer	ar_bounded + kmeans_8	73	0.16 [0.14, 0.19]	0.90	+23	0.074	-0.06 [-0.09, -0.03]	yes	2 / 0 / 2 of 10
		3	LSTM	no_ar + kmeans_8	86	0.14 [0.11, 0.16]	0.91	+77	0.078	-0.08 [-0.11, -0.06]	yes	0 / 0 / 0 of 10
0.01	40%	1	Pareto/NBD	—	66	0.25 [0.23, 0.28]	0.68	+34	—	—	—	—
		2	Transformer	ar_bounded + kmeans_8	102	0.16 [0.13, 0.18]	0.80	+64	0.062	-0.10 [-0.13, -0.06]	yes	4 / 4 / 0 of 10
		3	LSTM	no_ar + kmeans_8	140	0.14 [0.11, 0.16]	0.81	+136	0.067	-0.12 [-0.15, -0.08]	yes	0 / 0 / 0 of 10
0.01	60%	1	Pareto/NBD	—	88	0.26 [0.23, 0.30]	0.57	+41	—	—	—	—
		2	Transformer	ar_bounded + kmeans_8	103	0.18 [0.12, 0.23]	0.63	+43	0.047	-0.09 [-0.12, -0.05]	yes	5 / 5 / 1 of 10
		3	Transformer	no_ar + kmeans_8	120	0.16 [0.12, 0.21]	0.64	+65	0.047	-0.10 [-0.15, -0.05]	yes	4 / 2 / 0 of 10

Rate	Churn	Rank	Model	Arm	MAPE	Spearman	RMSE	Bias %	Val. CE	Δ vs rank 1 [95% CI]	Clearly behind rank 1	Beats P/NBD: MAPE / bias / Spearman
0.01	80%	1	Pareto/NBD	—	159	0.20 [0.16, 0.24]	0.42	+54	—	—	—	—
		2	Transformer	no_ar + kmeans_8	195	0.15 [0.12, 0.18]	0.48	+104	0.030	-0.05 [-0.07, -0.02]	yes	6 / 5 / 2 of 10
		3	LSTM	no_ar + kmeans_8	729	0.11 [0.07, 0.16]	0.85	+725	0.033	-0.09 [-0.12, -0.05]	yes	0 / 0 / 1 of 10
0.05	20%	1	Pareto/NBD	—	35	0.58 [0.56, 0.60]	1.92	+8	—	—	—	—
		2	Transformer	ar_unbounded	91	0.52 [0.50, 0.54]	2.69	+90	0.173	-0.06 [-0.07, -0.04]	yes	1 / 0 / 0 of 10
		3	Transformer	ar_bounded + kmeans_8	79	0.51 [0.48, 0.53]	2.65	+77	0.168	-0.07 [-0.09, -0.06]	yes	0 / 0 / 0 of 10
0.05	40%	1	Pareto/NBD	—	39	0.60 [0.58, 0.61]	1.63	+15	—	—	—	—
		2	Transformer	ar_bounded	65	0.55 [0.52, 0.57]	1.91	+53	0.138	-0.05 [-0.06, -0.03]	yes	3 / 2 / 0 of 10
		3	LSTM	ar_bounded	56	0.54 [0.51, 0.56]	1.88	+52	0.138	-0.06 [-0.08, -0.05]	yes	1 / 0 / 0 of 10
0.05	60%	1	Pareto/NBD	—	42	0.56 [0.55, 0.57]	1.32	+9	—	—	—	—
		2	LSTM	ar_bounded	67	0.52 [0.51, 0.53]	1.43	+60	0.100	-0.04 [-0.05, -0.03]	yes	1 / 1 / 0 of 10
		3	Transformer	ar_bounded	77	0.52 [0.50, 0.53]	1.47	+66	0.100	-0.05 [-0.05, -0.04]	yes	2 / 0 / 0 of 10
0.05	80%	1	Pareto/NBD	—	58	0.44 [0.42, 0.47]	0.89	+13	—	—	—	—
		2	Transformer	ar_bounded	102	0.40 [0.38, 0.44]	1.04	+76	0.061	-0.03 [-0.05, -0.02]	yes	2 / 1 / 1 of 10
		3	LSTM	ar_bounded	60	0.40 [0.37, 0.43]	0.97	+19	0.061	-0.04 [-0.05, -0.03]	yes	7 / 4 / 1 of 10
0.10	20%	1	Pareto/NBD	—	31	0.72 [0.71, 0.73]	2.91	+3	—	—	—	—

Rate	Churn	Rank	Model	Arm	MAPE	Spearman	RMSE	Bias %	Val. CE	Δ vs rank 1 [95% CI]	Clearly behind rank 1	Beats P/NBD: MAPE / bias / Spearman
		2	LSTM	ar_bounded	32	0.69 [0.68, 0.70]	3.32	+29	0.261	-0.03 [-0.04, -0.02]	yes	4 / 0 / 0 of 10
		3	Transformer	ar_bounded	52	0.68 [0.67, 0.69]	3.74	+43	0.262	-0.05 [-0.05, -0.04]	yes	4 / 0 / 0 of 10
0.10	40%	1	Pareto/NBD	—	32	0.73 [0.72, 0.74]	2.47	+1	—	—	—	—
		2	LSTM	ar_bounded	34	0.69 [0.68, 0.71]	2.79	+28	0.198	-0.03 [-0.04, -0.03]	yes	6 / 1 / 0 of 10
		3	Transformer	ar_bounded + kmeans_8	58	0.65 [0.64, 0.66]	3.42	+51	0.192	-0.08 [-0.09, -0.06]	yes	1 / 0 / 0 of 10
0.10	60%	1	Pareto/NBD	—	37	0.65 [0.64, 0.66]	2.00	+3	—	—	—	—
		2	LSTM	ar_bounded	33	0.63 [0.61, 0.64]	2.12	+20	0.140	-0.03 [-0.04, -0.02]	yes	7 / 1 / 0 of 10
		3	Transformer	ar_bounded	115	0.61 [0.59, 0.62]	2.74	+104	0.140	-0.05 [-0.06, -0.03]	yes	1 / 0 / 0 of 10
0.10	80%	1	Pareto/NBD	—	49	0.51 [0.49, 0.52]	1.37	+7	—	—	—	—
		2	LSTM	ar_bounded	45	0.47 [0.45, 0.49]	1.45	+4	0.077	-0.04 [-0.05, -0.03]	yes	9 / 8 / 0 of 10
		3	Transformer	ar_bounded	102	0.47 [0.45, 0.48]	1.57	+82	0.078	-0.04 [-0.05, -0.03]	yes	3 / 2 / 0 of 10
0.30	20%	1	Pareto/NBD	—	30	0.86 [0.85, 0.87]	6.93	-14	—	—	—	—
		2	LSTM	ar_bounded	12	0.85 [0.85, 0.86]	6.49	+10	0.493	-0.01 [-0.01, 0]	yes	10 / 8 / 0 of 10
		3	LSTM	no_ar	19	0.85 [0.85, 0.86]	6.92	+17	0.497	-0.01 [-0.01, 0]	yes	10 / 2 / 0 of 10
0.30	40%	1	Pareto/NBD	—	31	0.83 [0.82, 0.84]	5.99	-14	—	—	—	—
		2	LSTM	ar_bounded	13	0.82 [0.81, 0.82]	5.66	+2	0.355	-0.01 [-0.02, -0.01]	yes	10 / 9 / 0 of 10

Rate	Churn	Rank	Model	Arm	MAPE	Spearman	RMSE	Bias %	Val. CE	Δ vs rank 1 [95% CI]	Clearly behind rank 1	Beats P/NBD: MAPE / bias / Spearman
		3	LSTM	no_ar	14	0.81 [0.81, 0.82]	5.74	+7	0.361	-0.02 [-0.02, -0.01]	yes	10 / 8 / 0 of 10
0.30	60%	1	Pareto/NBD	—	32	0.77 [0.76, 0.77]	4.72	-12	—	—	—	—
		2	LSTM	ar_bounded	19	0.70 [0.69, 0.71]	4.60	-7	0.232	-0.07 [-0.08, -0.06]	yes	10 / 6 / 0 of 10
		3	LSTM	no_ar	18	0.70 [0.69, 0.71]	4.73	-1	0.237	-0.07 [-0.08, -0.06]	yes	10 / 7 / 0 of 10
0.30	80%	1	Pareto/NBD	—	35	0.70 [0.68, 0.72]	3.03	-12	—	—	—	—
		2	LSTM	ar_bounded	35	0.51 [0.48, 0.53]	3.01	-13	0.117	-0.19 [-0.21, -0.18]	yes	5 / 4 / 0 of 10
		3	LSTM	no_ar	32	0.51 [0.49, 0.53]	3.26	-15	0.121	-0.19 [-0.20, -0.18]	yes	7 / 3 / 0 of 10

Reading

- MAPE leader by cell.** Pareto/NBD leads all 8 cells at rates 0.01–0.05 and the two low-churn cells at rate 0.10. LSTM `ar_bounded` leads at rate 0.10 with churn 60–80% and at rate 0.30 with churn 20–40%; LSTM `no_ar` leads at rate 0.30 with churn 60–80%. We can also see that at lower transaction rate, the transformer performs better than the LSTM
- Most MAPE leads are not separated.** The runner-up is clearly behind the leader (paired interval excluding 0, 10 panels) in only 7 of 16 cells: behind Pareto/NBD at rate 0.05 with churn 20–60%, behind the LSTM at rate 0.10 with churn 60–80% (Pareto/NBD, by 4 points), and LSTM `no_ar` behind LSTM `ar_bounded` at rate 0.30 with churn 20–40%. At rate 0.01 no Transformer runner-up is clearly behind Pareto/NBD.
- Spearman leader by cell.** Pareto/NBD leads all 16 cells, and the runner-up is clearly behind it in every one. From rate 0.05 up the runner-up is almost always an `ar_bounded` arm. On sparse panels it is a `kmeans_8` arm: the cluster label raises the neural Spearman (LSTM `no_ar` 0.00 → 0.14 at rate 0.01, churn pooled) while leaving bias and MAPE no better.
- Wins on each metric.** A neural tree beats Pareto/NBD on Spearman in 14 of 1,918 panel comparisons: 12 on rate-0.01 panels, where both rank customers at only 0.1–0.3, and 2 at rate 0.05, churn 80%, by less than 0.01. Wins on |bias| are rarer than wins on MAPE: at rate 0.30 LSTM `ar_bounded` beats Pareto/NBD's MAPE on 35 of 40 panels but its |bias| on 27, and at rate 0.10, churn 60%, on 7 of 10 against 1. A lower MAPE there comes from the weekly shape, not the yearly level. These counts are description; the tests are in claim 1.
- Good MAPE does not mean good ranking.** At rate 0.30 and churn 80% the LSTM leads on MAPE (32) but ranks customers at 0.51 against Pareto/NBD's 0.70. The paired Spearman gap between Pareto/NBD and the best neural arm (`ar_bounded`) is supported in all 16 cells for both models (claim 1): 0.26–0.34 (LSTM) and 0.12–0.27 (Transformer) at rate 0.01, and 0.01–0.22 at rates 0.05–0.30.

- **Validation CE.** The CE gaps between arms (third decimal) are small beside the spread across panels, so the intervals overlap throughout. Within one panel, a lower CE across the 12 neural arms goes with a better holdout. Each panel gives one rank correlation over its 12 arms, tested against 0 inside each cell (table below): with MAPE it is supported and positive in 15 of 16 cells (cell means +0.21 to +0.70), with RMSE in 15 (+0.19 to +0.66) and with Spearman negative in 15 (−0.20 to −0.60). The one exception to all three is rate 0.01, churn 20%, where no correlation is clear. The ordering is loose at the top: the `kmeans_8` arms often have a lower CE and a worse MAPE than their `no_cluster` counterparts.

Impact of cohort size

Tripling the cohort from 1,000 to 3,000 customers halves the LSTM's bias and MAPE on sparse and mid-rate panels. The gain is largest with bounded flags. Pareto/NBD improves on MAPE only.

Setting. Seasonal panels (4 peaks, amplitude 1.5), 156 weeks, 1,000 vs 3,000 generated customers. The 3,000 grid is identical otherwise: same rates, churn levels, seed and training budget. The Transformer was not run at 3,000. Each entry is RMSE on customer totals / bias % / MAPE, mean over 40 panels. The entries pool four cells, so they are description only and carry no marks; the per-cell tests are claim 12 below. A 1,000 row and a 3,000 row are different customer populations, so they are compared as independent replications.

By purchase rate (the 4 churn levels pooled):

Model	Customers	Rate 0.01	0.05	0.10	0.30
Pareto/NBD	1,000	0.62 / +35 / 90	1.44 / +11 / 44	2.19 / +4 / 37	5.17 / −13 / 32
Pareto/NBD	3,000	0.61 / +19 / 56	1.44 / +7 / 34	2.17 / 0 / 32	5.12 / −13 / 30
LSTM <code>no_ar</code>	1,000	0.84 / +309 / 314	2.10 / +199 / 200	2.71 / +93 / 98	5.16 / +2 / 21
LSTM <code>no_ar</code>	3,000	0.78 / +219 / 219	1.78 / +117 / 119	2.35 / +5 / 24	4.83 / −5 / 14
LSTM <code>ar_bounded</code>	1,000	0.77 / +231 / 243	1.64 / +44 / 58	2.42 / +20 / 36	4.94 / −2 / 20
LSTM <code>ar_bounded</code>	3,000	0.64 / +65 / 83	1.53 / +18 / 31	2.25 / +1 / 18	4.77 / −11 / 16

By churn (the 4 rates pooled; RMSE here mixes rates, so compare it within a column only):

Model	Customers	Churn 20%	40%	60%	80%
Pareto/NBD	1,000	3.14 / +2 / 36	2.69 / +9 / 42	2.15 / +10 / 50	1.43 / +16 / 75
Pareto/NBD	3,000	3.11 / +1 / 32	2.66 / +5 / 35	2.11 / +5 / 39	1.44 / +1 / 46
LSTM <code>no_ar</code>	1,000	3.35 / +37 / 42	2.97 / +74 / 77	2.49 / +127 / 134	2.01 / +365 / 380
LSTM <code>no_ar</code>	3,000	3.05 / +33 / 34	2.74 / +57 / 59	2.26 / +89 / 99	1.69 / +156 / 184
LSTM <code>ar_bounded</code>	1,000	3.24 / +32 / 39	2.78 / +49 / 56	2.22 / +66 / 79	1.54 / +147 / 183
LSTM <code>ar_bounded</code>	3,000	2.96 / +15 / 21	2.60 / +15 / 26	2.12 / +20 / 39	1.51 / +24 / 63

Tests (claim 12: per cell, 10 vs 10 panels, independent bootstrap):

- **LSTM `ar_bounded`** : MAPE improves in all 12 cells at rates 0.01–0.10 and |bias| in 10 of them (not at churn 80% of rates 0.05 and 0.10). At rate 0.30 MAPE improves at churn 20–40% only (−5 and −3 points) and |bias| at churn 20% only.
- **LSTM `no_ar`** : MAPE and |bias| improve in every cell at rates 0.05 and 0.10, and at rate 0.30 with churn 20–60% (MAPE). At rate 0.01 the result is mixed: better at churn 80% (−360 MAPE points), worse at churn 20% (+13), no clear difference at churn 40–60%.
- **Pareto/NBD**: MAPE improves in 14 of 16 cells, all except churn 20–40% at rate 0.30, and |bias| in 7, all at rates 0.01–0.10. There is no clear RMSE difference in any cell.

Reading. Cohort size matters most where each customer buys least. The bounded flags need data to learn what silence means: at 1,000 customers they cut the LSTM's bias at churn 80% from +365 to +147; at 3,000, from +156 to +24. The churn trend in the neural bias is partly a cohort-size effect: at 1,000 generated customers, churn 80% panels keep only about 510 active customers against about 800 at churn 20%.

Impact of AR features

Bounded flags help both models, most on mid-rate panels (0.05–0.10) and at high churn, and barely at rate 0.30 (one LSTM cell on |bias|). Unbounded counters make the LSTM several times worse at every rate, and hurt the Transformer only on dense panels.

Setting. Seasonal panels, 1,000 customers, no cluster label. `no_ar` = count and calendar only; `ar_bounded` = 0/1 flags for activity in the last 2 / 4 / 8 / 16 / 32 weeks plus has-bought-before; `ar_unbounded` = recency, frequency and age as counters. Each cell is RMSE on customer totals / bias % / MAPE, mean over 40 panels. The entries pool four cells, so they are description only and carry no marks; the tests are per cell, below.

-

Reading.

- The flags matter where silence is informative but the model cannot learn it alone: mid-rate panels. On the sparsest panels silence says little; on the densest the plain LSTM already gets the level right (bias +2%).
- Unbounded counters keep growing through the holdout, past any value seen in training, and the LSTM's forecast climbs with them. Its RMSE rises from 2.1–5.2 to 11.9–16.3 at rates 0.05–0.30.
- Even with flags, the neural bias at churn 80% (+147 LSTM, +73 Transformer) stays far above Pareto/NBD's +16 at 1,000 customers. The cohort-size section shows that gap mostly closes for the LSTM at 3,000.

Claims and whether the evidence supports them

Every claim below is tested per rate × churn cell and, where the data allow, at both cohort sizes. Rates, churn levels and the whole grid are then described by the pattern of those per-cell verdicts. Most verdicts depend on the conditions, so each claim states where it holds and where it does not. All numbers are recomputed from the stored forecasts by `.scratch/synthetic-grid/metrics_by_cell.py` and `claims.py`.

Setting. Seasonal panels (4 peaks, amplitude 1.5), 1,000 customers unless a table says 3,000. Four purchase rates × four churn rates (20, 40, 60 and 80% of customers dropped out by week 52), 10 panels per cell, 40 per rate, 160 over the grid.

How to read the tests (`docs/statistical-protocol.md`).

- **Effect Δ .** Every comparison is written "A → B" and $\Delta = \text{mean}(B) - \text{mean}(A)$ over the cell's panels. For MAPE, |bias| and RMSE a negative Δ means B is better; for Spearman, `R_A` and shape correlation a positive Δ does. MAPE and |bias| are in percentage points.
- **Interval.** The 95% percentile-bootstrap interval of Δ from 10,000 resamples (`panelclv.evaluation.effects.effect`). **Paired** comparisons (two trees on the same panels) resample the panels with their pairs intact. **Independent** comparisons (1,000 vs 3,000 customers, claim 12) resample each side separately.
- **Bold** means supported: the cell's interval excludes 0. An interval containing 0 means no clear difference at $n = 10$, never "no difference".
- **Pooled entries are description.** In each grid, the "churn pooled" column and the "all" row (in italics) are pooled mean differences over 40 or 160 panels, with no interval and no verdict. The collapsed "by rate" tables give the pooled means and, per metric, how many

of the rate's cells support a difference and in which direction ("3/4 (–)" = three of four cells, all with B lower).

- **Grids.** One grid per model and context: rows are purchase rates, columns churn levels, and each cell reads " Δ MAPE / Δ |bias| / Δ Spearman". The intervals behind every cell are in the collapsed tables under each grid. An interval end printed as 0 is not exactly 0 before rounding.

Summary

#	Claim	Verdict	Holds	Does not hold
1	Pareto/NBD beats the neural models.	Partly	Ranking (Spearman): 94 of 96 cell comparisons, both cohort sizes. Level (MAPE): against the Transformer in 12–14 of 16 cells and never the other way; against the LSTM in every cell at rate 0.01 and at rate 0.05 with churn 20–60%.	LSTM <code>ar_bounded</code> on MAPE at rate 0.10 (churn 60–80%) and rate 0.30 (churn 20–60%) at 1,000 customers, and at rates 0.10–0.30 at 3,000; the LSTM ranks slightly better (+0.003) at 3,000, rate 0.30, churn 20%.
2	Neural error rises with churn.	Partly (described, not tested)	bias of the flagless LSTM at rates 0.01–0.05, both cohort sizes; MAPE of most trees.	Not neural-specific for MAPE: Pareto/NBD's also rises at every step at all four rates. With the flags the LSTM's bias no longer rises steadily at rates 0.05–0.10.
3	Bounded AR flags help.	Supported, conditionally	LSTM: 10 of 16 cells on MAPE, at rates 0.01–0.10 from churn 40–60% up (–24 to –437 points per supported cell); at 3,000 customers every cell at rate 0.01. Transformer: 6 of 16 cells, mostly at rates 0.05–0.10.	Churn 20% below rate 0.30; rate 0.30, where the LSTM changes by a few points either way; most Transformer cells.
4	Unbounded counters hurt.	Supported for the LSTM, partly for the Transformer	LSTM: 13 of 16 cells, every rate, +38 to +818 MAPE points. Transformer: 5 cells, all at rates 0.10–0.30.	Transformer at rates 0.01–0.05; ranking improves for both models at rate 0.05.
5	A k-means cluster label hurts.	Partly	LSTM level from rate 0.10 up (from 0.05 with flags).	It raises Spearman on sparse panels; with the unbounded counters it helps the Transformer and mostly helps the LSTM, except two rate-0.01 cells where it hurts badly.
6	One architecture is better overall.	Not supported	—	The Transformer is better at rate 0.01 and with unbounded counters; the LSTM at rates 0.10–0.30 with <code>no_ar</code> or flags.
7	Neural models capture seasonality; Pareto/NBD cannot.	Supported	Shape correlation 0.11–0.96 (LSTM) against –0.09 to +0.13 (Pareto/NBD); the true season cuts Pareto/NBD's MAPE by 5–20 points, supported in every cell.	—
8	Pareto/NBD's low bias means it is accurate per customer.	Not supported	—	In every cell $R_A < 1$ and $L_D > 0$ are both supported: it serves living customers only 51–87% of their volume and leaks 11–104% onto dead ones.
9	Neural models cannot detect a customer who has stopped.	Partly	Transformer: more leakage than Pareto/NBD in 15 of 16 cells. LSTM: every cell at rates 0.01–0.10, and rate 0.30 with churn 20–40%.	LSTM <code>ar_bounded</code> at rate 0.30 with churn 60–80%: no clear difference from Pareto/NBD.

#	Claim	Verdict	Holds	Does not hold
10	A bigger hyperparameter search helps.	Partly	Level at rate 0.30 (3 LSTM cells, 2 Transformer); the Transformer's ranking in 11 of 16 cells.	Level at rates 0.01–0.10 (a few scattered cells, one of them worse).
11	Specific hyperparameters drive the error.	Not supported (described, not tested)	Only as a pattern: the Transformer no_ar layer count correlates positively with bias and MAPE in 13 of 16 cells.	Within cells nearly every hyperparameter's correlation splits in sign across cells; the large pooled correlations are the regime choosing both variables.
12	More customers improve the neural forecast.	Supported, conditionally	LSTM ar_bounded : every cell at rates 0.01–0.10. LSTM no_ar : every cell at rates 0.05–0.10, most at 0.30.	LSTM no_ar at rate 0.01 (mixed: worse at churn 20%, better at 80%); LSTM ar_bounded at rate 0.30 above churn 40%. The Transformer was not run at 3,000.
13	RMSE can rank these models.	Not supported for per-week RMSE; supported within one rate for customer-total RMSE (described, not tested)	Customer-total RMSE ranks trees like MAPE (ρ +0.88 to +0.97 within each rate).	Per-week RMSE puts 4–10 of 13 trees on the same value to two decimals.

1. Pareto/NBD beats the neural models

Verdict: partly. Pareto/NBD ranks customers better than the neural tree in 94 of the 96 cell comparisons (16 cells × six neural contexts). The other two are both LSTM arms at 3,000 customers, rate 0.30, churn 20%, which rank customers better than Pareto/NBD by +0.003 and +0.004: supported, and small. On the level of the forecast Pareto/NBD beats the Transformer wherever a difference is clear and the LSTM on sparse panels, but the LSTM beats it on MAPE on dense panels, and at 3,000 customers from rate 0.10 up.

- **Ranking.** Δ Spearman is supported in all 96 cells, negative in 94. The gap is largest at rate 0.01 (−0.12 to −0.34) and at rate 0.30 with churn 80% (−0.19 to −0.26).
- **Level, LSTM ar_bounded .** Pareto/NBD is better on MAPE in every cell at rate 0.01 (+16 to +433 points for the LSTM) and at rate 0.05 with churn 20–60% (+13 to +24); churn 80% at rate 0.05 shows no clear difference. At rate 0.10 the LSTM is better on MAPE at churn 60–80% (by 4 points each), while |bias| favours Pareto/NBD at churn 20–60% (+12 to +27). At rate 0.30 with churn 20–60% the LSTM is better by 13–18 MAPE points, and by 5–8 |bias| points at churn 20–40%.
- **Level, Transformer.** Pareto/NBD is better on MAPE in 12 of 16 cells against the Transformer with flags and 14 of 16 without, and the Transformer is better in none. The cells with no clear difference are rate 0.01 with churn 40–80% and rate 0.30 with churn 20% (with flags), and rate 0.30 with churn 20–40% (without).
- **At 3,000 customers** the LSTM ar_bounded beats Pareto/NBD on MAPE in every cell at rate 0.10 (−4 to −19 points) and at rate 0.30 with churn 20–60% (−13 to −24). At rate 0.05 no cell shows a clear difference, and at rate 0.01 Pareto/NBD is better at churn 40–80%. The LSTM still has the larger |bias| in 12 of 16 cells.

Δ = neural minus Pareto/NBD: positive MAPE/|bias| or negative Spearman means Pareto/NBD is better.

LSTM ar_bounded , 1,000 customers: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+16 / +34 / −0.26	+55 / +80 / −0.34	+108 / +149 / −0.31	+433 / +521 / −0.27	+153 / +196 / −0.29
0.05	+13 / +38 / −0.07	+17 / +37 / −0.06	+24 / +48 / −0.04	+2 / +6 / −0.04	+14 / +32 / −0.06
0.10	+1 / +27 / −0.03	+3 / +23 / −0.03	−4 / +12 / −0.03	−4 / −5 / −0.04	−1 / +14 / −0.03

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.30	-18 / -5 / -0.01	-18 / -8 / -0.01	-13 / -2 / -0.07	0 / +6 / -0.19	-12 / -2 / -0.07
all					+38 / +60 / -0.11

LSTM no_ar , 1,000 customers: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+23 / +47 / -0.21	+71 / +98 / -0.24	+203 / +249 / -0.25	+600 / +701 / -0.22	+224 / +274 / -0.23
0.05	+9 / +33 / -0.29	+60 / +83 / -0.40	+115 / +145 / -0.53	+439 / +477 / -0.44	+156 / +184 / -0.42
0.10	+2 / +28 / -0.05	+26 / +51 / -0.09	+32 / +52 / -0.05	+184 / +204 / -0.24	+61 / +84 / -0.11
0.30	-11 / +3 / -0.01	-16 / -5 / -0.02	-14 / -4 / -0.07	-3 / +4 / -0.19	-11 / 0 / -0.07
all					+108 / +135 / -0.21

Transformer ar_bounded , 1,000 customers: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+14 / +32 / -0.19	+10 / +24 / -0.27	+10 / +20 / -0.20	+17 / +35 / -0.12	+13 / +28 / -0.20
0.05	+33 / +54 / -0.12	+26 / +39 / -0.05	+35 / +56 / -0.05	+44 / +64 / -0.03	+34 / +53 / -0.06
0.10	+21 / +42 / -0.05	+9 / +28 / -0.09	+78 / +104 / -0.05	+53 / +69 / -0.04	+40 / +61 / -0.05
0.30	+7 / +12 / -0.05	+11 / +21 / -0.13	+35 / +49 / -0.10	+19 / +33 / -0.22	+18 / +29 / -0.12
all					+26 / +43 / -0.11

Transformer no_ar , 1,000 customers: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+10 / +25 / -0.17	+31 / +42 / -0.24	+41 / +61 / -0.20	+38 / +67 / -0.13	+30 / +49 / -0.18
0.05	+39 / +63 / -0.27	+71 / +93 / -0.18	+107 / +131 / -0.19	+64 / +91 / -0.12	+70 / +95 / -0.19
0.10	+27 / +53 / -0.14	+62 / +85 / -0.15	+66 / +90 / -0.13	+216 / +244 / -0.10	+93 / +118 / -0.13
0.30	+6 / +10 / -0.15	+5 / +15 / -0.16	+25 / +37 / -0.18	+61 / +75 / -0.26	+24 / +34 / -0.19
all					+54 / +74 / -0.17

LSTM ar_bounded , 3,000 customers: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+1 / +14 / -0.18	+5 / +15 / -0.21	+27 / +43 / -0.19	+74 / +105 / -0.22	+27 / +44 / -0.20
0.05	-5 / +17 / -0.04	-4 / +15 / -0.05	-1 / +15 / -0.04	-2 / +8 / -0.04	-3 / +14 / -0.04
0.10	-17 / +9 / -0.02	-19 / +4 / -0.01	-15 / +4 / -0.01	-4 / +7 / -0.04	-14 / +6 / -0.02
0.30	-24 / -12 / 0	-20 / -7 / -0.01	-13 / +1 / -0.07	0 / +12 / -0.19	-14 / -1 / -0.07
all					-1 / +16 / -0.08

LSTM no_ar , 3,000 customers: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+47 / +73 / -0.22	+85 / +109 / -0.29	+199 / +231 / -0.28	+322 / +377 / -0.28	+163 / +197 / -0.27
0.05	-5 / +18 / -0.07	+36 / +59 / -0.13	+72 / +95 / -0.23	+238 / +265 / -0.36	+85 / +109 / -0.20

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.10	−13 / +15 / −0.04	−2 / +26 / −0.04	−15 / +5 / −0.02	−3 / +15 / −0.04	−8 / +15 / −0.03
0.30	−23 / −9 / 0	−21 / −10 / −0.01	−18 / −6 / −0.07	−3 / +11 / −0.19	−16 / −3 / −0.07
all					+56 / +80 / −0.14

- By rate, churn pooled (descriptive)
- Intervals per rate × churn cell

2. Neural error rises with churn

Verdict: partly, as a described pattern. A trend across churn levels runs across cells, and the protocol tests only within a cell, so this claim is described, not tested.

MAPE rises at every churn step for most trees, Pareto/NBD included (at all four rates at 1,000 customers), so that half of the claim is not specific to the neural models. What is specific is the size of the |bias| rise without flags: the LSTM no_ar goes from 59 to 755 at rate 0.01 and from 42 to 497 at rate 0.05, against 12 to 54 and 8 to 21 for Pareto/NBD. With bounded flags the LSTM's |bias| no longer rises steadily at rates 0.05–0.10, at either cohort size. It still rises at rate 0.01 and, at 3,000 customers, at rate 0.30 (3 → 24).

Each row gives the mean at churn 20 / 40 / 60 / 80% over the level's 10 panels, and whether the mean rises at every step. RMSE is left out: it falls with churn for every tree because there is less volume.

Tree	Customers	Rate	bias at churn 20 / 40 / 60 / 80%	MAPE at churn 20 / 40 / 60 / 80%	bias rises at every step	MAPE rises at every step
Pareto/NBD	1,000	0.01	12 / 34 / 41 / 54	48 / 66 / 88 / 159	yes	yes
Pareto/NBD	1,000	0.05	8 / 15 / 12 / 21	35 / 39 / 42 / 58	no	yes
Pareto/NBD	1,000	0.10	3 / 5 / 9 / 21	31 / 32 / 37 / 49	yes	yes
Pareto/NBD	1,000	0.30	14 / 14 / 13 / 13	30 / 31 / 32 / 35	no	yes
LSTM no_ar	1,000	0.01	59 / 132 / 290 / 755	71 / 136 / 291 / 759	yes	yes
LSTM no_ar	1,000	0.05	42 / 98 / 157 / 497	45 / 99 / 158 / 497	yes	yes
LSTM no_ar	1,000	0.10	31 / 57 / 61 / 225	33 / 58 / 69 / 233	yes	yes
LSTM no_ar	1,000	0.30	17 / 9 / 9 / 17	19 / 14 / 18 / 32	no	no
LSTM ar_bounded	1,000	0.01	47 / 114 / 190 / 576	65 / 121 / 196 / 591	yes	yes
LSTM ar_bounded	1,000	0.05	47 / 52 / 61 / 27	48 / 56 / 67 / 60	no	no
LSTM ar_bounded	1,000	0.10	30 / 28 / 21 / 15	32 / 34 / 33 / 45	no	no
LSTM ar_bounded	1,000	0.30	10 / 6 / 10 / 19	12 / 13 / 19 / 35	no	yes
Transformer no_ar	1,000	0.01	37 / 76 / 102 / 121	58 / 96 / 129 / 197	yes	yes
Transformer no_ar	1,000	0.05	72 / 108 / 144 / 111	75 / 110 / 149 / 122	no	no
Transformer no_ar	1,000	0.10	56 / 91 / 99 / 265	58 / 94 / 103 / 265	yes	yes
Transformer no_ar	1,000	0.30	24 / 29 / 50 / 88	36 / 36 / 57 / 96	yes	no
Transformer ar_bounded	1,000	0.01	44 / 58 / 60 / 90	62 / 75 / 98 / 176	yes	yes
Transformer ar_bounded	1,000	0.05	63 / 54 / 69 / 84	68 / 65 / 77 / 102	no	no

Tree	Customers	Rate	bias at churn 20 / 40 / 60 / 80%	MAPE at churn 20 / 40 / 60 / 80%	bias rises at every step	MAPE rises at every step
Transformer ar_bounded	1,000	0.10	45 / 33 / 113 / 90	52 / 41 / 115 / 102	no	no
Transformer ar_bounded	1,000	0.30	27 / 35 / 62 / 46	37 / 42 / 67 / 54	no	no
Pareto/NBD	3,000	0.01	11 / 22 / 30 / 22	37 / 46 / 62 / 78	no	yes
Pareto/NBD	3,000	0.05	7 / 10 / 7 / 8	32 / 33 / 34 / 39	no	yes
Pareto/NBD	3,000	0.10	2 / 2 / 3 / 7	30 / 30 / 32 / 35	no	yes
Pareto/NBD	3,000	0.30	15 / 14 / 13 / 12	30 / 30 / 30 / 31	no	no
LSTM no_ar	3,000	0.01	83 / 131 / 261 / 399	84 / 132 / 261 / 399	yes	yes
LSTM no_ar	3,000	0.05	25 / 69 / 102 / 274	26 / 69 / 105 / 277	yes	yes
LSTM no_ar	3,000	0.10	17 / 27 / 8 / 22	18 / 29 / 17 / 32	no	no
LSTM no_ar	3,000	0.30	6 / 4 / 7 / 23	8 / 9 / 12 / 28	no	yes
LSTM ar_bounded	3,000	0.01	25 / 37 / 72 / 127	38 / 52 / 89 / 152	yes	yes
LSTM ar_bounded	3,000	0.05	24 / 25 / 22 / 16	26 / 29 / 33 / 37	no	yes
LSTM ar_bounded	3,000	0.10	11 / 6 / 7 / 14	14 / 12 / 16 / 32	no	no
LSTM ar_bounded	3,000	0.30	3 / 7 / 13 / 24	6 / 10 / 18 / 31	yes	yes

3. Bounded AR flags help

Verdict: supported, conditionally. The flags improve the LSTM's level at purchase rates 0.01–0.10 once churn is 40% or more (60% or more at rate 0.01), and its ranking in every cell at rates 0.05–0.10. The gain grows with churn, is absent at churn 20% below rate 0.30, and is larger with more customers. At rate 0.30 the effect is small and mixed. For the Transformer they help less: the level improves in 6 of 16 cells, the ranking in 10.

How much, LSTM at 1,000 customers.

- MAPE improves in 10 of 16 cells: at rate 0.01 with churn 60–80% (–95 and –168 points), at rates 0.05 and 0.10 with churn 40–80% (–24 to –437), and at rate 0.30 with churn 20–40% (–8 and –2). It worsens in one, rate 0.30 with churn 80% (+3). |bias| improves in 9 cells, the same pattern. Over the grid the pooled mean MAPE falls from 158 to 89.
- By churn: at rate 0.05 the change is +4 (no clear difference), –43, –91 and –437 MAPE points at churn 20, 40, 60 and 80%. The same rise with churn holds at rates 0.01 and 0.10. Where silence is common, the flags are what lets the model learn that silence means dropout.
- Ranking improves in all eight cells at rates 0.05–0.10 (+0.02 to +0.49 Spearman) but worsens in two cells at rate 0.01 (–0.10 and –0.05) and does not move at rate 0.30.

How much, Transformer at 1,000 customers.

- MAPE improves in 6 of 16 cells, by 31–163 points: rate 0.01 at churn 60%, rate 0.05 at churn 40–60%, rate 0.10 at churn 40% and 80%, rate 0.30 at churn 80%. No cell worsens. Over the grid the pooled mean MAPE falls from 105 to 77.
- Ranking improves in 10 cells, every cell at rate 0.05 and three of four at rates 0.10 and 0.30, by 0.04–0.15. It worsens slightly in one (rate 0.01, churn 20%, –0.02).

How much, LSTM at 3,000 customers.

- The flags help more on sparse panels: MAPE improves in every cell at rate 0.01 (−46 to −248) and at rate 0.05 with churn 40–80% (−40 to −240). The pooled mean falls from 94 to 37.
- At 3,000 customers the flags also improve the ranking at rate 0.01 (three of four cells, +0.04 to +0.09), where they worsened it at 1,000.
- At rate 0.30 they cost a little: MAPE +6 at churn 60%, and |bias| +3 and +7 at churn 40–60%.

LSTM, 1,000 customers: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	−6 / −13 / −0.05	−16 / −17 / −0.10	−95 / −100 / −0.05	−168 / −179 / −0.05	−71 / −77 / −0.06
0.05	+4 / +5 / +0.21	−43 / −46 / +0.34	−91 / −97 / +0.49	−437 / −470 / +0.40	−142 / −152 / +0.36
0.10	0 / −1 / +0.02	−24 / −28 / +0.06	−36 / −40 / +0.03	−188 / −210 / +0.20	−62 / −70 / +0.08
0.30	−8 / −8 / 0	−2 / −3 / 0	+1 / +1 / 0	+3 / +2 / 0	−1 / −2 / 0
all					−69 / −75 / +0.09

Transformer, 1,000 customers: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+4 / +7 / −0.02	−21 / −18 / −0.04	−31 / −41 / 0	−21 / −31 / +0.01	−17 / −21 / −0.01
0.05	−6 / −9 / +0.14	−45 / −54 / +0.13	−72 / −75 / +0.15	−20 / −27 / +0.09	−36 / −41 / +0.13
0.10	−6 / −11 / +0.10	−53 / −58 / +0.06	+12 / +14 / +0.08	−163 / −175 / +0.06	−53 / −57 / +0.08
0.30	+1 / +2 / +0.09	+6 / +6 / +0.03	+10 / +13 / +0.08	−42 / −42 / +0.04	−6 / −5 / +0.06
all					−28 / −31 / +0.06

LSTM, 3,000 customers: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	−46 / −59 / +0.04	−80 / −94 / +0.09	−172 / −189 / +0.09	−248 / −271 / +0.06	−136 / −153 / +0.07
0.05	0 / −1 / +0.03	−40 / −44 / +0.07	−73 / −81 / +0.19	−240 / −258 / +0.33	−88 / −96 / +0.16
0.10	−4 / −6 / +0.02	−17 / −22 / +0.02	−1 / −1 / 0	−1 / −8 / 0	−6 / −9 / +0.01
0.30	−1 / −2 / 0	+1 / +3 / 0	+6 / +7 / 0	+2 / +1 / 0	+2 / +2 / 0
all					−57 / −64 / +0.06

By rate, churn pooled (descriptive)

Comparison	Rate	n (A / B)	MAPE A → B	bias A → B	Δ MAPE (pooled mean) · cells supported	Δ bias (pooled mean) · cells supported	Δ RMSE (pooled mean) · cells supported	Δ Spearman (pooled mean) · cells supported
LSTM, 1,000 customers	0.01	40 / 40	314 → 243	309 → 232	−71 · 2/4 (−)	−77 · 2/4 (−)	−0.07 · 2/4 (−)	−0.06 · 2/4 (−)
LSTM, 1,000 customers	0.05	40 / 40	200 → 58	199 → 46	−142 · 3/4 (−)	−152 · 3/4 (−)	−0.46 · 3/4 (−)	+0.36 · 4/4 (+)
LSTM, 1,000 customers	0.10	40 / 40	98 → 36	93 → 24	−62 · 3/4 (−)	−70 · 3/4 (−)	−0.29 · 3/4 (−)	+0.08 · 4/4 (+)
LSTM, 1,000 customers	0.30	40 / 40	21 → 20	13 → 11	−1 · 3/4 (1+, 2−)	−2 · 1/4 (−)	−0.22 · 3/4 (−)	0 · 0/4

Comparison	Rate	n (A / B)	MAPE A → B	bias A → B	Δ MAPE (pooled mean) · cells supported	Δ bias (pooled mean) · cells supported	Δ RMSE (pooled mean) · cells supported	Δ Spearman (pooled mean) · cells supported
LSTM, 1,000 customers	all	160 / 160	158 → 89	154 → 78	-69 · 11/16 (1+, 10-)	-75 · 9/16 (-)	-0.26 · 11/16 (-)	+0.09 · 10/16 (8+, 2-)
Transformer, 1,000 customers	0.01	40 / 40	120 → 103	84 → 63	-17 · 1/4 (-)	-21 · 0/4	-0.02 · 1/4 (-)	-0.01 · 1/4 (-)
Transformer, 1,000 customers	0.05	40 / 40	114 → 78	109 → 68	-36 · 2/4 (-)	-41 · 2/4 (-)	-0.25 · 3/4 (-)	+0.13 · 4/4 (+)
Transformer, 1,000 customers	0.10	40 / 40	130 → 77	128 → 70	-53 · 2/4 (-)	-57 · 2/4 (-)	-0.45 · 2/4 (-)	+0.08 · 3/4 (+)
Transformer, 1,000 customers	0.30	40 / 40	56 → 50	48 → 42	-6 · 1/4 (-)	-5 · 1/4 (-)	-0.49 · 1/4 (-)	+0.06 · 3/4 (+)
Transformer, 1,000 customers	all	160 / 160	105 → 77	92 → 61	-28 · 6/16 (-)	-31 · 5/16 (-)	-0.30 · 7/16 (-)	+0.06 · 11/16 (10+, 1-)
LSTM, 3,000 customers	0.01	40 / 40	219 → 83	219 → 65	-136 · 4/4 (-)	-153 · 4/4 (-)	-0.14 · 4/4 (-)	+0.07 · 3/4 (+)
LSTM, 3,000 customers	0.05	40 / 40	119 → 31	118 → 22	-88 · 3/4 (-)	-96 · 3/4 (-)	-0.26 · 3/4 (-)	+0.16 · 4/4 (+)
LSTM, 3,000 customers	0.10	40 / 40	24 → 18	19 → 9	-6 · 2/4 (-)	-9 · 3/4 (-)	-0.10 · 4/4 (-)	+0.01 · 3/4 (2+, 1-)
LSTM, 3,000 customers	0.30	40 / 40	14 → 16	10 → 12	+2 · 1/4 (+)	+2 · 3/4 (2+, 1-)	-0.05 · 1/4 (-)	0 · 0/4
LSTM, 3,000 customers	all	160 / 160	94 → 37	91 → 27	-57 · 10/16 (1+, 9-)	-64 · 13/16 (2+, 11-)	-0.14 · 12/16 (-)	+0.06 · 10/16 (9+, 1-)

► Intervals per rate × churn cell

4. Unbounded counters hurt

Verdict: supported for the LSTM, partly for the Transformer. The counters make the LSTM's level worse in 13 of 16 cells, at every rate (+38 to +818 MAPE points per supported cell), because they keep growing through the holdout, past any value seen in training. The three cells with no clear difference are at rate 0.01 (churn 40% and 80%) and rate 0.10 (churn 80%), where the LSTM without counters is already far off. The Transformer is worse on MAPE in 5 cells, all at rates 0.10–0.30, and in none at rates 0.01–0.05. Ranking does not follow the level: the counters raise both models' Spearman at rate 0.05 (LSTM in all four cells, +0.15 to +0.32; Transformer in three, +0.10 to +0.21).

LSTM: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+443 / +443 / -0.17	+52 / +50 / -0.24	+223 / +224 / -0.21	+79 / +66 / -0.10	+199 / +196 / -0.18
0.05	+195 / +198 / +0.15	+214 / +215 / +0.19	+304 / +304 / +0.32	+818 / +817 / +0.27	+383 / +384 / +0.23
0.10	+268 / +269 / -0.07	+317 / +318 / -0.07	+239 / +240 / -0.08	+73 / +70 / +0.15	+224 / +224 / -0.02
0.30	+248 / +250 / -0.12	+147 / +149 / -0.08	+46 / +48 / -0.04	+38 / +39 / -0.03	+120 / +122 / -0.07
all					+231 / +231 / -0.01

Transformer: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	-9 / -18 / -0.10	-18 / -21 / -0.02	-2 / -22 / -0.01	+148 / +152 / -0.07	+30 / +23 / -0.05
0.05	+16 / +18 / +0.21	-6 / -8 / +0.10	-25 / -22 / +0.11	-22 / -35 / -0.12	-9 / -12 / +0.08
0.10	+43 / +45 / +0.07	-3 / -1 / +0.06	+56 / +60 / +0.05	-20 / -20 / +0.04	+19 / +21 / +0.05
0.30	+35 / +45 / +0.05	+49 / +53 / +0.01	+38 / +44 / +0.04	+14 / +15 / -0.01	+36 / +41 / +0.03
all					+19 / +19 / +0.02

- By rate, churn pooled (descriptive)
- Intervals per rate × churn cell

5. A k-means cluster label hurts

Verdict: partly. The label's effect depends on the model, the AR encoding and the rate.

- **LSTM no_ar** : no clear change in level at rate 0.01; one better cell at rate 0.05 (churn 40%, -17 MAPE); worse at rate 0.10 with churn 60–80% (+64, +76) and in every cell at rate 0.30 (+9 to +110). It raises Spearman in every cell at rates 0.01–0.05 (+0.12 to +0.35) and lowers it in six cells at rates 0.10–0.30.
- **LSTM ar_bounded** : worse level in 10 cells, most from rate 0.05 up (+14 to +229 MAPE per supported cell), better in two (rate 0.01 with churn 80%, -146; rate 0.05 with churn 20%, -9). Better ranking at rate 0.01 only (all four cells), worse in the other 12.
- **Transformer no_ar and ar_bounded** : the level changes clearly in only 2 and 4 of 16 cells (all four ar_bounded cells worse, +17 to +77). Spearman rises on sparse panels (seven of the eight rate-0.01 cells, +0.07 to +0.18).
- **With ar_unbounded** the label helps the Transformer in 8 cells (all at rates 0.05–0.30) and hurts it in 1. For the LSTM it helps in 5 cells at rates 0.05–0.30 but hurts in 2 at rate 0.01, by more than 1,000 MAPE points each, so its pooled grid mean even rises (+52). Where it helps, it helps by displacing the broken counters.

LSTM no_ar : $\Delta \text{MAPE} / \Delta |\text{bias}| / \Delta \text{Spearman}$

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+15 / +18 / +0.12	+3 / +4 / +0.13	-18 / -18 / +0.15	-30 / -30 / +0.13	-7 / -7 / +0.13
0.05	+2 / -1 / +0.17	-17 / -17 / +0.26	+15 / +15 / +0.35	+9 / +9 / +0.35	+2 / +1 / +0.28
0.10	+6 / +4 / -0.09	+17 / +17 / -0.07	+64 / +71 / -0.09	+76 / +84 / +0.11	+40 / +44 / -0.04
0.30	+9 / +10 / -0.11	+39 / +45 / -0.07	+72 / +81 / -0.06	+110 / +125 / -0.03	+57 / +65 / -0.07
all					+23 / +26 / +0.08

LSTM ar_bounded : $\Delta \text{MAPE} / \Delta |\text{bias}| / \Delta \text{Spearman}$

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+8 / +12 / +0.17	+1 / +1 / +0.20	+48 / +51 / +0.12	-146 / -147 / +0.14	-22 / -21 / +0.15
0.05	-9 / -18 / -0.05	+9 / +9 / -0.09	+43 / +47 / -0.09	+229 / +260 / -0.04	+68 / +75 / -0.07
0.10	0 / -2 / -0.09	+19 / +23 / -0.06	+60 / +71 / -0.09	+89 / +116 / -0.02	+42 / +52 / -0.07
0.30	+14 / +15 / -0.07	+32 / +38 / -0.06	+52 / +60 / -0.05	+49 / +64 / -0.02	+37 / +44 / -0.05
all					+31 / +38 / -0.01

LSTM ar_unbounded : $\Delta \text{MAPE} / \Delta |\text{bias}| / \Delta \text{Spearman}$

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	-264 / -268 / +0.07	+1115 / +1118 / +0.11	+390 / +386 / +0.16	+1177 / +1172 / +0.10	+605 / +602 / +0.11
0.05	-51 / -52 / +0.02	-84 / -92 / -0.01	-103 / -105 / +0.01	-395 / -400 / +0.05	-158 / -162 / +0.02
0.10	-156 / -159 / -0.02	-201 / -208 / -0.01	-165 / -168 / +0.03	-118 / -124 / +0.02	-160 / -165 / +0.01
0.30	-187 / -192 / +0.02	-99 / -108 / +0.01	-10 / -18 / 0	-23 / -30 / 0	-80 / -87 / +0.01
all					+52 / +47 / +0.04

Transformer no_ar : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	-3 / -2 / +0.07	+52 / +57 / +0.11	-9 / -15 / +0.10	-2 / -12 / +0.08	+10 / +7 / +0.09
0.05	-7 / -9 / +0.18	-3 / -4 / +0.06	-14 / -13 / +0.11	+59 / +56 / +0.04	+9 / +7 / +0.10
0.10	-1 / -2 / +0.03	+1 / +3 / +0.05	+6 / +8 / +0.04	-62 / -63 / +0.02	-14 / -14 / +0.03
0.30	-6 / -3 / +0.05	+19 / +24 / +0.03	+8 / +15 / +0.02	+23 / +31 / +0.01	+11 / +17 / +0.03
all					+4 / +4 / +0.06

Transformer ar_bounded : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+10 / +15 / +0.13	+27 / +22 / +0.18	+5 / +7 / +0.12	+20 / +23 / +0.03	+16 / +16 / +0.11
0.05	+11 / +14 / +0.05	-9 / -7 / -0.03	+77 / +83 / -0.05	-8 / -11 / -0.03	+18 / +20 / -0.01
0.10	-7 / -7 / -0.03	+17 / +21 / +0.01	-6 / -7 / -0.04	+75 / +86 / -0.05	+20 / +23 / -0.03
0.30	-5 / -2 / -0.05	+5 / +10 / +0.04	+17 / +21 / -0.02	+36 / +41 / -0.01	+13 / +17 / -0.01
all					+17 / +19 / +0.02

Transformer ar_unbounded : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+16 / +27 / +0.08	+5 / +5 / 0	-7 / +4 / +0.04	-130 / -128 / +0.09	-29 / -23 / +0.05
0.05	-22 / -25 / -0.05	-25 / -33 / -0.05	-21 / -24 / -0.05	-3 / -4 / +0.19	-18 / -22 / +0.01
0.10	-55 / -62 / -0.02	-14 / -17 / -0.01	-83 / -89 / -0.01	-94 / -103 / -0.01	-62 / -68 / -0.01
0.30	-34 / -40 / +0.03	-45 / -54 / +0.05	-35 / -45 / +0.01	-60 / -67 / +0.03	-42 / -50 / +0.03
all					-38 / -41 / +0.02

- By rate, churn pooled (descriptive)
- Intervals per rate × churn cell

6. One architecture is better overall

Verdict: not supported. Neither architecture is better across the grid: in every arm the cells split, with the Transformer clearly better in some and the LSTM clearly better in others. The pooled grid means (Transformer minus LSTM: -53 MAPE with no_ar , -12 with flags) are dominated by the LSTM's blow-ups at rate 0.01 and describe no typical cell.

- **Rate 0.01:** the Transformer is better on MAPE in three of four cells with no_ar and with flags (-13 to -562 points) and in all four with the counters, and ranks customers better in seven of the eight no_ar and ar_bounded cells.

- **Rates 0.10–0.30:** the LSTM is better on MAPE in 7 of 8 cells with `no_ar` and 7 of 8 with flags (+17 to +82 for the Transformer), and ranks customers better in every cell at rate 0.30.
- **With unbounded counters** the Transformer is better in 12 of 16 cells, every cell at rates 0.01–0.05: it tolerates the counters and the LSTM does not. At rate 0.30 the advantage is gone (one cell each way).

Δ = Transformer minus LSTM on the same panels.

`no_ar` : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	-13 / -23 / +0.04	-40 / -55 / +0.01	-162 / -188 / +0.06	-562 / -634 / +0.09	-194 / -225 / +0.05
0.05	+30 / +30 / +0.02	+11 / +10 / +0.22	-9 / -13 / +0.34	-375 / -386 / +0.32	-86 / -90 / +0.23
0.10	+25 / +25 / -0.09	+36 / +34 / -0.06	+34 / +38 / -0.07	+32 / +40 / +0.13	+32 / +34 / -0.02
0.30	+17 / +7 / -0.14	+22 / +20 / -0.14	+39 / +41 / -0.11	+64 / +71 / -0.07	+35 / +35 / -0.11
all					-53 / -62 / +0.03

`no_ar` + `kmeans_8` : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	-31 / -42 / -0.01	+9 / -2 / -0.01	-153 / -184 / 0	-534 / -616 / +0.04	-177 / -211 / 0
0.05	+21 / +22 / +0.03	+24 / +23 / +0.02	-38 / -41 / +0.10	-325 / -339 / +0.01	-79 / -84 / +0.04
0.10	+19 / +19 / +0.03	+20 / +19 / +0.06	-24 / -26 / +0.06	-105 / -107 / +0.05	-23 / -24 / +0.05
0.30	+2 / -6 / +0.02	+1 / 0 / -0.04	-24 / -25 / -0.03	-22 / -23 / -0.02	-11 / -14 / -0.02
all					-73 / -83 / +0.02

`ar_bounded` : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	-2 / -2 / +0.06	-45 / -56 / +0.07	-98 / -129 / +0.10	-416 / -486 / +0.15	-140 / -169 / +0.10
0.05	+20 / +16 / -0.05	+9 / +2 / +0.01	+11 / +8 / 0	+42 / +57 / +0.01	+20 / +21 / -0.01
0.10	+20 / +15 / -0.02	+6 / +5 / -0.05	+82 / +92 / -0.02	+57 / +74 / 0	+41 / +46 / -0.02
0.30	+26 / +17 / -0.05	+29 / +29 / -0.11	+48 / +52 / -0.04	+19 / +27 / -0.02	+31 / +31 / -0.05
all					-12 / -17 / 0

`ar_bounded` + `kmeans_8` : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+1 / 0 / +0.03	-19 / -36 / +0.05	-142 / -174 / +0.10	-249 / -316 / +0.04	-102 / -131 / +0.06
0.05	+40 / +48 / +0.06	-8 / -13 / +0.07	+44 / +44 / +0.04	-195 / -213 / +0.01	-30 / -34 / +0.04
0.10	+12 / +10 / +0.04	+4 / +3 / +0.01	+15 / +14 / +0.04	+43 / +44 / -0.03	+19 / +18 / +0.02
0.30	+7 / 0 / -0.03	+2 / +1 / -0.01	+13 / +13 / -0.01	+7 / +4 / -0.02	+7 / +4 / -0.02
all					-27 / -36 / +0.02

`ar_unbounded` : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	−464 / −484 / +0.11	−111 / −127 / +0.23	−387 / −434 / +0.26	−493 / −549 / +0.12	−364 / −398 / +0.18
0.05	−149 / −150 / +0.08	−210 / −212 / +0.13	−337 / −340 / +0.13	−1215 / −1237 / −0.07	−478 / −485 / +0.07
0.10	−199 / −200 / +0.05	−284 / −285 / +0.06	−148 / −142 / +0.05	−62 / −50 / +0.03	−173 / −169 / +0.05
0.30	−196 / −198 / +0.03	−76 / −77 / −0.05	+31 / +36 / −0.03	+53 / +64 / −0.04	−48 / −46 / −0.02
all					−265 / −274 / +0.07

ar_unbounded + kmeans_8 : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	−185 / −189 / +0.11	−1221 / −1240 / +0.11	−784 / −817 / +0.14	−1799 / −1849 / +0.11	−997 / −1023 / +0.12
0.05	−120 / −124 / +0.01	−151 / −153 / +0.09	−256 / −259 / +0.07	−823 / −842 / +0.06	−337 / −345 / +0.06
0.10	−98 / −102 / +0.05	−97 / −95 / +0.06	−65 / −63 / +0.01	−39 / −29 / −0.01	−75 / −72 / +0.03
0.30	−43 / −46 / +0.04	−21 / −22 / −0.01	+12 / +16 / −0.02	+9 / +17 / −0.02	−11 / −9 / 0
all					−355 / −362 / +0.05

- By rate, churn pooled (descriptive)
- Intervals per rate × churn cell

7. Neural models capture seasonality; Pareto/NBD cannot

Verdict: supported for this one seasonal pattern.

- **Shape.** The LSTM's predicted weekly totals correlate with the actual ones at 0.11–0.96 and the Transformer's at 0.09–0.62, against −0.09 to +0.13 for Pareto/NBD. The correlation rises with the rate and falls with churn.
- **Counterfactual.** Multiplying Pareto/NBD's forecast by the true seasonal curve (rescaled to mean 1) lowers its MAPE by 5–20 points, supported in all 16 cells. |bias| and Spearman barely move (at most 1.0 point and 0.001 per cell): several of those tiny changes are supported, because the rescaling moves every panel by nearly the same amount, but the season changes the weekly shape, not the yearly total or the order of customers.
- **Given the season, Pareto/NBD beats the LSTM** on MAPE in every cell at rates 0.01–0.10 (+7 to +439 points for the LSTM) and at rate 0.30 with churn 80% (+12). At rate 0.30 with churn 20–40% the LSTM keeps a small lead (2–3 points), and churn 60% shows no clear difference. So the LSTM's dense-panel MAPE wins in claim 1 come mostly from modelling the season.

Shape correlation, mean over panels: Pareto/NBD / LSTM ar_bounded / Transformer ar_bounded

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled
0.01	−0.02 / +0.26 / +0.32	0 / +0.29 / +0.22	−0.02 / +0.14 / +0.11	+0.05 / +0.11 / +0.09	0 / +0.20 / +0.19
0.05	−0.09 / +0.70 / +0.50	+0.04 / +0.64 / +0.51	+0.06 / +0.53 / +0.35	+0.06 / +0.21 / +0.23	+0.02 / +0.52 / +0.40
0.10	−0.07 / +0.84 / +0.62	+0.01 / +0.73 / +0.50	+0.10 / +0.67 / +0.41	+0.11 / +0.36 / +0.30	+0.04 / +0.65 / +0.46
0.30	−0.08 / +0.96 / +0.60	+0.01 / +0.91 / +0.55	+0.12 / +0.83 / +0.53	+0.13 / +0.49 / +0.47	+0.05 / +0.80 / +0.54

Pareto/NBD → with true season: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	−9 / 0 / 0	−8 / 0 / 0	−5 / −1 / 0	−6 / −1 / 0	−7 / −1 / 0
0.05	−20 / 0 / 0	−14 / −1 / 0	−14 / −1 / 0	−7 / −1 / 0	−14 / −1 / 0

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.10	−20 / 0 / 0	−18 / 0 / 0	−16 / 0 / 0	−11 / 0 / 0	−16 / 0 / 0
0.30	−15 / 0 / 0	−16 / +1 / 0	−16 / +1 / 0	−12 / +1 / 0	−15 / +1 / 0
all					−13 / 0 / 0

Pareto/NBD with season → LSTM ar_bounded : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+25 / +35 / −0.25	+63 / +81 / −0.34	+113 / +150 / −0.31	+439 / +522 / −0.27	+160 / +197 / −0.29
0.05	+33 / +38 / −0.07	+31 / +37 / −0.06	+38 / +49 / −0.04	+9 / +7 / −0.04	+28 / +33 / −0.06
0.10	+22 / +27 / −0.03	+21 / +23 / −0.03	+12 / +13 / −0.03	+7 / −5 / −0.04	+15 / +15 / −0.03
0.30	−3 / −5 / −0.01	−2 / −8 / −0.01	+3 / −3 / −0.07	+12 / +5 / −0.19	+2 / −3 / −0.07
all					+51 / +60 / −0.11

- By rate, churn pooled (descriptive)
- Intervals per rate × churn cell

8. Pareto/NBD's low bias means it is accurate per customer

Verdict: not supported. In every cell Pareto/NBD under-serves customers who are still alive (alive ratio R_A 0.51–0.87; R_A < 1 supported in all 16 cells) and assigns volume to customers who have already died (dead leakage L_D 0.11–1.04; L_D > 0 supported in all 16). Each panel gives one R_A and one L_D, tested per cell against 1 and 0 by the one-statistic rule, paired, 10 panels a cell. The two errors cancel in the total, which is why its |bias| is small. Both worsen with churn: at rate 0.01 and churn 80% it puts more volume on dead customers (L_D 1.04) than the living ones should get. R_A and L_D use the generator's hidden truth (each customer's λ and death week).

Rate	Churn	n	bias (mean)	R_A: mean, R_A − 1 [95% CI]	L_D: mean [95% CI]	RMSE (mean)
0.01	20%	10	12	0.85, −0.15 [−0.18, −0.12]	0.28 [0.27, 0.28]	0.79
0.01	40%	10	34	0.77, −0.23 [−0.26, −0.20]	0.55 [0.52, 0.59]	0.68
0.01	60%	10	41	0.65, −0.35 [−0.39, −0.30]	0.76 [0.72, 0.80]	0.57
0.01	80%	10	54	0.51, −0.49 [−0.53, −0.45]	1.04 [0.90, 1.21]	0.42
0.05	20%	10	8	0.87, −0.13 [−0.15, −0.12]	0.22 [0.21, 0.24]	1.92
0.05	40%	10	15	0.77, −0.23 [−0.25, −0.21]	0.39 [0.38, 0.40]	1.63
0.05	60%	10	12	0.65, −0.35 [−0.37, −0.32]	0.44 [0.40, 0.48]	1.32
0.05	80%	10	21	0.56, −0.44 [−0.49, −0.39]	0.53 [0.46, 0.60]	0.89
0.10	20%	10	3	0.84, −0.16 [−0.17, −0.15]	0.18 [0.17, 0.19]	2.91
0.10	40%	10	5	0.74, −0.26 [−0.28, −0.24]	0.27 [0.26, 0.29]	2.47
0.10	60%	10	9	0.67, −0.33 [−0.35, −0.30]	0.35 [0.31, 0.40]	2.00
0.10	80%	10	21	0.60, −0.40 [−0.44, −0.36]	0.42 [0.35, 0.49]	1.37
0.30	20%	10	14	0.74, −0.26 [−0.26, −0.25]	0.11 [0.11, 0.12]	6.93
0.30	40%	10	14	0.68, −0.32 [−0.33, −0.31]	0.19 [0.17, 0.20]	5.99
0.30	60%	10	13	0.63, −0.37 [−0.38, −0.35]	0.25 [0.23, 0.27]	4.72
0.30	80%	10	13	0.58, −0.42 [−0.44, −0.39]	0.29 [0.24, 0.34]	3.03

Rate	Churn	n	bias (mean)	R_A: mean, R_A - 1 [95% CI]	L_D: mean [95% CI]	RMSE (mean)
0.01	pooled (descriptive)	40	35	0.70	0.66	0.62
0.05	pooled (descriptive)	40	14	0.71	0.40	1.44
0.10	pooled (descriptive)	40	9	0.71	0.31	2.19
0.30	pooled (descriptive)	40	13	0.66	0.21	5.17
all	pooled (descriptive)	160	18	0.70	0.39	2.35

9. Neural models cannot detect a customer who has stopped

Verdict: partly. On dead leakage L_D, lower is better.

- **LSTM ar_bounded** leaks more than Pareto/NBD in every cell at rates 0.01–0.10 (+0.10 to +4.44), most at rate 0.01, churn 80%, and slightly more at rate 0.30 with churn 20–40% (+0.05, +0.03). At rate 0.30 with churn 60–80% there is no clear difference (+0.01 and -0.02 at n = 10): there the flags come closest to working.
- **Transformer ar_bounded** leaks more than Pareto/NBD in every cell (+0.08 to +0.83), supported in 15 of 16 (all but rate 0.01, churn 40%).

LSTM: $\Delta L_D / \Delta R_A$

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+0.14 / +0.18	+0.54 / +0.26	+1.25 / +0.29	+4.44 / +0.52	+1.59 / +0.31
0.05	+0.20 / +0.18	+0.34 / +0.03	+0.48 / +0.04	+0.27 / -0.21	+0.32 / +0.01
0.10	+0.14 / +0.12	+0.23 / +0.04	+0.17 / -0.01	+0.10 / -0.13	+0.16 / +0.01
0.30	+0.05 / +0.19	+0.03 / +0.12	+0.01 / +0.05	-0.02 / +0.01	+0.02 / +0.09
all					+0.52 / +0.11

Transformer: $\Delta L_D / \Delta R_A$

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+0.11 / +0.12	+0.08 / -0.16	+0.28 / -0.14	+0.55 / -0.13	+0.25 / -0.08
0.05	+0.29 / +0.24	+0.36 / +0.03	+0.58 / 0	+0.83 / -0.19	+0.51 / +0.02
0.10	+0.21 / +0.20	+0.26 / 0	+0.79 / +0.20	+0.65 / +0.06	+0.48 / +0.11
0.30	+0.12 / +0.27	+0.19 / +0.18	+0.37 / +0.35	+0.37 / +0.21	+0.26 / +0.25
all					+0.38 / +0.08

- By rate, churn pooled (descriptive)
- Intervals per rate × churn cell

10. A bigger hyperparameter search helps

Verdict: partly. From the archived 10-trial (LSTM) and 20-trial (Transformer) no_ar runs on the same panels to the 100-trial runs, the level improves most consistently at rate 0.30: in three of four cells for the LSTM (churn 40–80%, -19 to -42 MAPE) and two for the Transformer (churn 40–60%, -22 and -39). Below rate 0.30 the level changes clearly in only a few scattered cells: two for the LSTM (both better) and four for the Transformer (three better, one worse: rate 0.01, churn 60%, +38). The ranking improves in 11 of 16 cells for the Transformer, every cell at rate 0.30, three of four at rates 0.05 and 0.10 and one at rate 0.01 (+0.03 to +0.22), and in 5 for the LSTM (all four at rate 0.30 and one

at rate 0.10, by 0.01–0.03). The archived run did not record its embedder, so the two runs may also differ in that.

LSTM, 10 → 100 trials: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	−3 / −4 / 0	−49 / −53 / +0.01	−32 / −32 / 0	+47 / +45 / −0.01	−9 / −11 / 0
0.05	−10 / −9 / −0.13	+6 / +6 / −0.02	−32 / −33 / −0.06	+79 / +79 / 0	+11 / +11 / −0.05
0.10	−7 / −5 / +0.03	−11 / −12 / −0.02	−32 / −39 / 0	−46 / −52 / −0.04	−24 / −27 / −0.01
0.30	−3 / −2 / +0.03	−19 / −23 / +0.02	−42 / −46 / +0.01	−30 / −38 / +0.01	−23 / −27 / +0.02
all					−12 / −14 / −0.01

Transformer, 20 → 100 trials: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	−14 / −23 / +0.02	+12 / +9 / −0.02	+38 / +55 / +0.02	−12 / −24 / +0.05	+6 / +4 / +0.02
0.05	−1 / −2 / +0.12	−41 / −42 / +0.22	−19 / −23 / +0.14	−143 / −149 / +0.22	−51 / −54 / +0.17
0.10	−19 / −20 / +0.03	−7 / −8 / +0.05	−24 / −25 / +0.08	−20 / −19 / +0.04	−17 / −18 / +0.05
0.30	+2 / −6 / +0.04	−22 / −26 / +0.04	−39 / −45 / +0.03	−69 / −72 / +0.03	−32 / −37 / +0.04
all					−24 / −26 / +0.07

- By rate, churn pooled (descriptive)
- Intervals per rate × churn cell

11. Specific hyperparameters drive the error

Verdict: not supported; described only, no test. A correlation between a chosen hyperparameter and the error, over a cell's 10 panels, is not a difference of means, so the protocol's interval cannot be put on it; and pooling the 16 cells into one test is not allowed. The table therefore describes the correlations. Pooled over the grid, they mix cells, and there the rate and churn choose both the hyperparameter and the error: the LSTM `no_ar` batch size correlates with |bias| at +0.48 over the grid, but within cells it is positive in 5 of 12 cells and negative in 7. Within cells almost every hyperparameter splits roughly evenly between positive and negative, with cell values spread from about −0.7 to +0.7. The one consistent pattern is the Transformer `no_ar` layer count: positive in 13 of 16 cells against both |bias| and MAPE (more encoder layers with a worse level). The LSTM `ar_bounded` dense width leans negative (10 of 15 cells). Neither pattern is a tested effect.

Tree	Hyperparameter	ρ with bias , pooled	ρ with bias within cells: positive / negative of cells, range	ρ with MAPE, pooled	ρ with MAPE within cells: positive / negative of cells, range
LSTM <code>no_ar</code>	<code>batch_size</code>	+0.48	5 / 7 of 12, −0.63 to +0.62	+0.49	5 / 5 of 12, −0.60 to +0.62
LSTM <code>no_ar</code>	<code>learning_rate</code>	−0.15	9 / 7 of 16, −0.61 to +0.93	−0.16	8 / 8 of 16, −0.61 to +0.87
LSTM <code>no_ar</code>	<code>lstm_hidden_size</code>	−0.20	9 / 6 of 15, −0.66 to +0.64	−0.21	9 / 6 of 15, −0.66 to +0.57
LSTM <code>no_ar</code>	<code>dense_units</code>	+0.28	8 / 7 of 15, −0.73 to +0.57	+0.28	8 / 7 of 15, −0.78 to +0.43
LSTM <code>no_ar</code>	<code>dropout</code>	−0.05	12 / 4 of 16, −0.76 to +0.66	−0.05	11 / 5 of 16, −0.76 to +0.66
LSTM <code>ar_bounded</code>	<code>batch_size</code>	+0.07	7 / 6 of 13, −0.43 to +0.59	+0.06	6 / 6 of 13, −0.32 to +0.71

Tree	Hyperparameter	ρ with bias , pooled	ρ with bias within cells: positive / negative of cells, range	ρ with MAPE, pooled	ρ with MAPE within cells: positive / negative of cells, range
LSTM ar_bounded	learning_rate	-0.02	10 / 6 of 16, -0.44 to +0.47	-0.10	10 / 6 of 16, -0.39 to +0.32
LSTM ar_bounded	lstm_hidden_size	0	9 / 7 of 16, -0.69 to +0.54	-0.04	9 / 7 of 16, -0.45 to +0.62
LSTM ar_bounded	dense_units	-0.28	5 / 10 of 15, -0.85 to +0.51	-0.22	5 / 10 of 15, -0.62 to +0.49
LSTM ar_bounded	dropout	-0.02	12 / 4 of 16, -0.61 to +0.59	+0.03	11 / 5 of 16, -0.61 to +0.56
Transformer no_ar	batch_size	-0.04	5 / 8 of 13, -0.50 to +0.70	-0.03	5 / 8 of 13, -0.52 to +0.70
Transformer no_ar	learning_rate	-0.09	8 / 8 of 16, -0.54 to +0.54	+0.05	8 / 8 of 16, -0.54 to +0.57
Transformer no_ar	d_model	-0.01	1 / 1 of 2, -0.17 to +0.17	-0.03	1 / 1 of 2, -0.06 to +0.09
Transformer no_ar	nhead	+0.08	10 / 6 of 16, -0.76 to +0.66	+0.05	10 / 6 of 16, -0.72 to +0.78
Transformer no_ar	num_encoder_layers	+0.24	13 / 3 of 16, -0.29 to +0.57	+0.29	13 / 2 of 16, -0.05 to +0.57
Transformer no_ar	dropout	+0.17	10 / 6 of 16, -0.35 to +0.60	+0.13	10 / 6 of 16, -0.44 to +0.65
Transformer ar_bounded	batch_size	+0.03	11 / 5 of 16, -0.61 to +0.57	+0.04	11 / 5 of 16, -0.61 to +0.53
Transformer ar_bounded	learning_rate	-0.05	4 / 12 of 16, -0.75 to +0.52	+0.13	7 / 9 of 16, -0.70 to +0.58
Transformer ar_bounded	d_model	-0.13	0 / 1 of 1, -0.52 to -0.52	-0.13	0 / 1 of 1, -0.52 to -0.52
Transformer ar_bounded	nhead	-0.05	8 / 7 of 16, -0.76 to +0.44	-0.01	8 / 7 of 16, -0.76 to +0.32
Transformer ar_bounded	num_encoder_layers	-0.01	8 / 8 of 16, -0.54 to +0.80	-0.10	9 / 7 of 16, -0.49 to +0.79
Transformer ar_bounded	dropout	-0.01	8 / 8 of 16, -0.70 to +0.62	+0.03	7 / 9 of 16, -0.66 to +0.68

12. More customers improve the neural forecast

Verdict: supported, conditionally. Tripling the cohort (1,000 $\rightarrow$ 3,000 customers).

Panel j of the two grids shares its seed, so its first 1,000 customers share their purchase-rate draw λ ; but the generator draws all N λ 's before any dropout rate μ , so the μ 's, the death weeks and every count come from a different part of the random stream (checked: the μ 's of the first 1,000 customers correlate at -0.01). The two grids are therefore different customer populations, and each cell is compared as 10 vs 10 independent replications, not paired.

- **LSTM ar_bounded** : MAPE improves in every cell at rates 0.01–0.10 (-13 to -439 points), and at rate 0.30 at churn 20–40% (-5, -3). Ranking improves in every cell at rate

0.01 (+0.07 to +0.17) and in one cell at each other rate.

- **LSTM no_ar** : MAPE improves in every cell at rates 0.05–0.10 and at rate 0.30 with churn 20–60%. At rate 0.01 it is mixed: better at churn 80% (–360), worse at churn 20% (+13), no clear difference at churn 40–60%. Without flags, extra customers do not reliably help the sparsest panels.
- **Pareto/NBD**: MAPE improves in 14 of 16 cells, modestly (–1 to –81 points, most at high churn); there is no clear RMSE difference in any cell.
- The Transformer was not run at 3,000 customers.

LSTM no_ar : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	+13 / +24 / –0.02	–5 / –1 / –0.02	–30 / –28 / –0.01	–360 / –356 / +0.03	–95 / –90 / 0
0.05	–18 / –17 / +0.22	–30 / –29 / +0.28	–52 / –55 / +0.31	–221 / –223 / +0.10	–80 / –81 / +0.23
0.10	–15 / –14 / +0.01	–29 / –29 / +0.05	–52 / –53 / +0.03	–201 / –203 / +0.21	–74 / –75 / +0.08
0.30	–12 / –12 / +0.01	–6 / –5 / +0.01	–6 / –2 / –0.01	–4 / +6 / 0	–7 / –3 / 0
all					–64 / –62 / +0.08

LSTM ar_bounded : Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	–27 / –22 / +0.07	–69 / –78 / +0.17	–107 / –117 / +0.13	–439 / –449 / +0.14	–160 / –166 / +0.13
0.05	–22 / –22 / +0.04	–26 / –27 / +0.01	–34 / –39 / +0.01	–23 / –11 / +0.02	–26 / –25 / +0.02
0.10	–19 / –19 / +0.02	–23 / –23 / +0.01	–16 / –14 / +0.01	–13 / –2 / +0.01	–18 / –14 / +0.01
0.30	–5 / –6 / +0.01	–3 / +1 / +0.01	–2 / +3 / –0.01	–5 / +5 / +0.01	–4 / +1 / 0
all					–52 / –51 / +0.04

Pareto/NBD: Δ MAPE / Δ |bias| / Δ Spearman

Rate	Churn 20%	Churn 40%	Churn 60%	Churn 80%	Churn pooled (descriptive)
0.01	–11 / –1 / –0.01	–19 / –13 / +0.03	–26 / –11 / +0.02	–81 / –32 / +0.09	–34 / –14 / +0.03
0.05	–4 / –1 / 0	–6 / –5 / 0	–9 / –5 / 0	–19 / –12 / +0.02	–9 / –6 / +0.01
0.10	–1 / –1 / 0	–2 / –3 / –0.01	–5 / –6 / 0	–14 / –13 / +0.01	–5 / –6 / 0
0.30	0 / +1 / 0	–1 / 0 / 0	–2 / 0 / 0	–4 / –1 / 0	–2 / 0 / 0
all					–13 / –7 / +0.01

► By rate, churn pooled (descriptive)

► Intervals per rate × churn cell

13. RMSE can rank these models

Verdict (described, not tested): not supported for per-week RMSE; supported within

one rate for customer-total RMSE. RMSE is a descriptive metric under the protocol, so this

claim rests on the spread of the 13 tree means, not on an interval. Per customer-week, RMSE puts 10 of 13 trees on the same value to two decimals at rates 0.01–0.05, because almost every cell is a zero. On customer totals RMSE separates

the trees and orders them almost as MAPE does (p over the 13 trees +0.88 to +0.97). It grows

about 8-fold from rate 0.01 to 0.30, though, so it cannot be pooled across rates.

Rate	Per-week RMSE, range over trees	Trees at the modal per-week RMSE (2 dp)	Customer-total RMSE, range	ρ (customer-total RMSE, MAPE) over trees
0.01	0.071–0.209	10 of 13	0.62–7.05	+0.97
0.05	0.135–0.309	10 of 13	1.44–11.88	+0.90
0.10	0.181–0.351	7 of 13	2.19–12.71	+0.88
0.30	0.310–0.487	4 of 13	4.94–16.34	+0.88

What this does not show

- **Run-to-run noise.** Each panel has one study, so a cell's spread mixes panel variation with unseeded training variation. The per-cell intervals therefore measure robustness to drawing a new panel and training on it once, together; neither part can be separated.
- **Transformer at 3,000 customers.** Not run, so claim 12 is LSTM-only.
- **Other encodings.** Only none / bounded / unbounded history and $K = 8$ were tested. The flag bins {2, 4, 8, 16, 32} were not tuned, and the projected embedder was not run.
- **Changing seasonality.** It is fixed across the grid, so nothing here says how the models behave under other seasonal patterns.
- **Transfer to real data.** The generator is a Pareto/NBD, so the benchmark is correct by construction here. On real panels there is no known ceiling (see docs/benchmarks-real-panels.md).